Information-Geometric Port-Hamiltonian Learning for Adaptive B-Spline Deformation
Problem Formulation, Proposed Framework, and Current Implementation

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Abstract

We study deformation of an explicitly parameterized periodic B-spline toward a target shape observation as an operator-constrained inverse problem on a finite-dimensional configuration manifold. The stored control vertices and knot logits are coordinate variables: the positive fixed-period knot intervals lie in the interior of a simplex, while regular and simple curves form admissible open subsets of the parameter space. A spline-evaluation map sends a configuration to a parameterized curve in the ambient space $\mathbb{R}^d$, and quotienting by orientation-preserving reparameterizations—and, when appropriate, rigid pose—defines the corresponding shape class. The Hamiltonian state therefore belongs to the cotangent phase space $T^*\mathcal{Q}$, with (p_q, p_γ) interpreted as cotangent variables rather than additional ambient points. The current PyTorch implementation realizes this geometry in one coordinate chart and uses a constant diagonal coordinate metric.

The observed curve is sampled as an empirical measure and compared with a target measure by a finite-iteration Sinkhorn divergence or by alternative point-cloud discrepancies. These discrepancies define pullback potentials on the spline configuration manifold rather than intrinsic distances unless a compatible configuration or quotient metric is specified. We distinguish three information-geometric objects that act at different levels of the model. The task discrepancy supplies a scalar deformation potential. A measurement Fisher matrix $\mathcal{I}_{\text{meas}} = L^\top \Sigma_e^{-1} L$ measures local sensitivity of geometric-safety residuals to control-point and knot motion. A Souriau–Fisher matrix $\mathcal{I}_{\text{S}} = \text{Cov}[\mathbf{J}]$ measures ensemble sensitivity of the global SE(2) momentum map on the Lie-algebra parameter space. Planar control vertices remain points of $\mathbb{R}^2$ on which SE(2) acts collectively; the knot-interval vector belongs to a fixed-period open simplex and is invariant under that pose action. The learned port-Hamiltonian vector field is represented in local cotangent coordinates, with skew interconnection and positive-semidefinite dissipation enforced by construction. The manuscript concludes with a reproducible audit protocol, a task-transparent deformation algorithm, and a precise separation between the intrinsic manifold formulation and the current chart-based implementation.

Contents

1 Introduction

Shape deformation is a representative operator-constrained inverse problem: given a source geometry, a target observation, and a family of admissible deformation dynamics, one seeks a trajectory that transforms the source into the target while preserving numerical stability and useful geometric structure. Explicit Lagrangian representations such as B-splines and NURBS provide compact degrees of freedom and exact boundary tracking, but their optimization is sensitive to point correspondences, parameterization, and local continuity constraints. Implicit level-set methods accommodate changes of interface topology more naturally, but they replace explicit boundary coordinates with an Eulerian field and can be costly when accurate boundary correspondence is required [1].

Optimal transport (OT) provides a principled way to compare empirical measures induced by source and target shapes. Entropic regularization makes the discrete problem differentiable and computationally tractable through Sinkhorn iterations [2], [3]. Debiased Sinkhorn divergences reduce the entropic self-bias and interpolate between transport-like and kernel-like behavior [4]. However, the transport potential is only one component of a deformation solver: a direct gradient flow may be over-damped, whereas conservative Hamiltonian dynamics may oscillate indefinitely.

Port-Hamiltonian systems separate conservative interconnection from dissipation and expose an energy balance suitable for structured learning and control [5], [6]. We use this structure to define a learned phase-space flow for

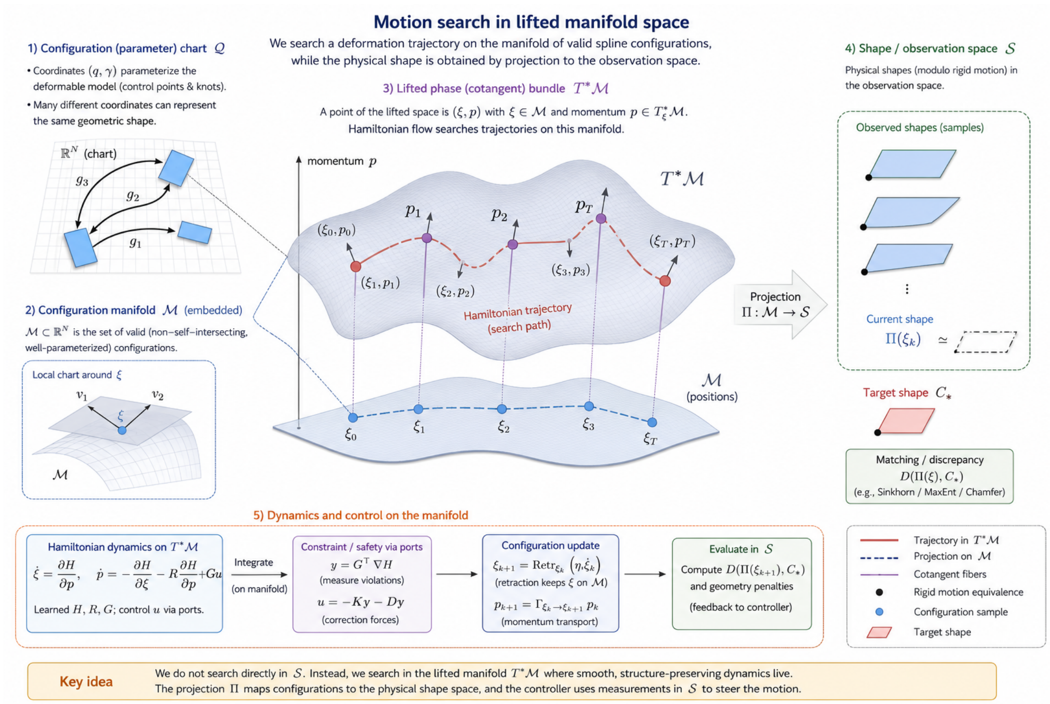


Figure 1: Remove the old improper statement and use a figure to explain the idea here. The motion we are searching lies in a configuration manifold for measurement-theoretical splines.

spline control points, latent knot variables, and their conjugate momenta. The present code initializes the learned interconnection near the canonical symplectic matrix and learns only residual interconnection and dissipation, which is substantially more stable than predicting an unconstrained vector field from scratch.

1.1 Problem statement, objectives, and contributions

Spline deformation problem on configuration and shape spaces (known target cases)

Let $\mathcal{Q}_{\mathrm{adm}}$ denote an admissible manifold of regular spline configurations, let $\mathcal{E} : \mathcal{Q}_{\mathrm{adm}} \to \mathrm{Emb}(S^1, \mathbb{R}^d)$ be the spline-evaluation map, and let $\pi_{\mathrm{sh}} : \mathrm{Emb}(S^1, \mathbb{R}^d) \to \mathcal{S}$ map a parameterized curve to its chosen shape class. Given an initial configuration $\xi_0 \in \mathcal{Q}_{\mathrm{adm}}$ and either a target class $[C_\star] \in \mathcal{S}$ or an observation Y, construct a cotangent-bundle trajectory

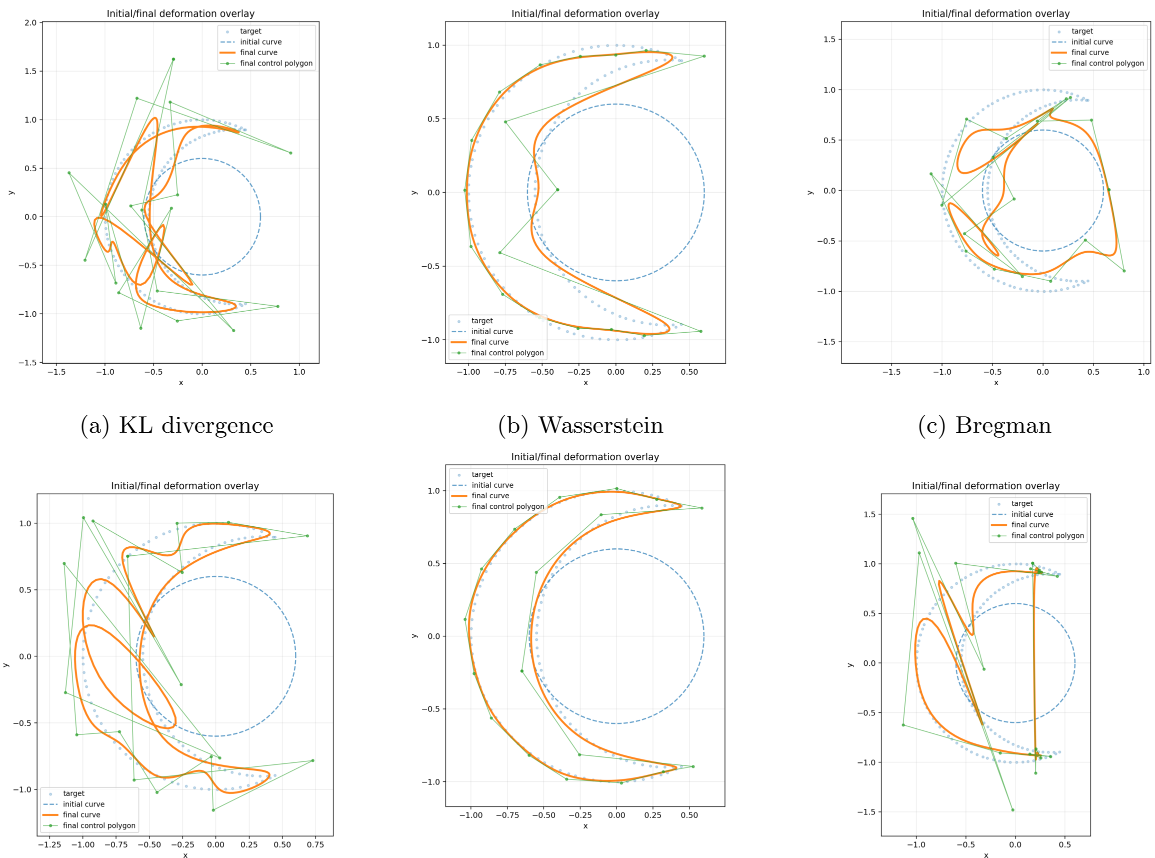
$$z(t) = (\xi(t), p(t)) \in T^*\mathcal{Q}_{\mathrm{adm}}, \qquad t \in [0, T],$$

with $\xi(0) = \xi_0$, such that $\pi_{\mathrm{sh}}(\mathcal{E}(\xi(T)))$ reaches or approaches the target class. When only Y is known, the terminal goal is the set

$$\mathcal{A}_Y^{\varepsilon} = \left\{ \xi \in \mathcal{Q}_{\mathrm{adm}} : D\big(\mathcal{O}(\xi), Y\big) \le \varepsilon \right\},$$

where $\mathcal{O}$ is a sampling, image, field, or measure-valued observation map. Temporal regularity and economy are measured on $\mathcal{Q}_{\mathrm{adm}}$ and $T^*\mathcal{Q}_{\mathrm{adm}}$, whereas geometric agreement is measured after applying $\mathcal{E}$ and, when required, the shape quotient π_{sh}.

The first objective is therefore not to learn an unconstrained pointwise displacement in the ambient plane, but to construct a path on an admissible configuration manifold whose image under spline evaluation approaches a target shape orbit. If one particular target parameterization is physically meaningful, a terminal condition can be imposed relative to a metric on $\mathcal{Q}_{\mathrm{adm}}$. If only the geometric shape is meaningful, the target is an equivalence class under reparameterization and possibly rigid pose, so a raw coordinate norm between control polygons or knot logits is not an invariant terminal loss. When only a point cloud, image boundary, or sampled geometry is available, the endpoint is specified by the inverse image of an observation-space tolerance set.



(a) KL divergence

(b) Wasserstein

(c) Bregman

(d) Max Entropy

(e) Sinkhorn Iteration (1 iters) as an approximation of Wasserstein

(f) Rényi Divergence of order 2

Figure 2: Terminal states of deformation with different divergence functionals as potential candidates in pH learning algorithms. All methods are trained under same scheme and evaluate at the same time horizon and currently no topological constraint are enforced as hard constraints in implementation. Optimal transport divergence is most stable and its entropy regularized version is the best performance in this circle to crescent task in particular. The guess is conservation of mass is automatically satisfied given the definition of optimal transport.

Our solution proceeds through three maps. First, the coordinate variables determine a configuration $\xi \in \mathcal{Q}_{\mathrm{adm}}$, and the evaluation/observation maps produce a curve, samples, or empirical measure $\mu_\xi = (\mathcal{E}(\xi))_{\#}\rho$. Information-geometric or optimal-transport discrepancies on the observation space then induce pullback potentials $V_Y = \mathcal{D}_Y \circ \mathcal{O}$ on $\mathcal{Q}_{\mathrm{adm}}$. Second, the configuration is lifted to its cotangent phase space $T^*\mathcal{Q}_{\mathrm{adm}}$, where a learned residual port-Hamiltonian template routes conservative motion through a skew bundle map and dissipates excess energy through a positive-semidefinite bundle map. Third, the pairwise problem is generalized conceptually to a distribution-to-distribution flow over configurations or shape classes, while feedback-driven active decomposition activates only the local coordinate–momentum blocks responsible for the unresolved mismatch.

The resulting contributions are:

1. **Configuration-to-configuration optimal control.** We state spline deformation literally as the problem of moving an initial control-point/knot configuration to a final configuration through a temporally regular trajectory, and we give both direct endpoint and observation-based terminal losses.
2. **Information-geometric hierarchy.** We separate the observation-space divergence used as a deformation potential, the configuration-level Fisher/Gauss–Newton pullback used for local shape and knot conditioning, and the Souriau–Fisher covariance metric used for global SE(2) momentum statistics. This prevents global pose conditioning from being conflated with local control-point motion or knot redistribution.
3. **Learned port-Hamiltonian solution template.** We formulate the deformation in phase space and learn residual interconnection and dissipation operators whose skew-symmetry and positive-semidefiniteness are enforced by construction, thereby exposing the energy and power balances used for training and diagnosis.
4. **Active decomposition and distributional extension.** We derive a passivity-preserving selector that updates complete control-point/knot coordinate–momentum pairs according to current geometric and transport residuals. This mechanism provides the building block for decomposing a population-level distribution-to-distribution flow into adaptive local deformation subflows; the population-level extension is a proposed generalization rather than a completed experimental result.
5. **Implementation audit.** We connect the mathematical stages to the current PyTorch pipeline, including finite-rollout training, Sinkhorn iteration effects, controller limitations, diagnostic quantities, and the separation between implemented, experimental, and future components.

2 Geometric representation and inverse problem

2.1 Spline configuration manifold and target shape quotient

Fix the spline degree, the number of periodic control vertices, and a total parameter period P. For a strict interval floor $\delta \geq 0$ satisfying $P > m\delta$, define the interval manifold

$$\mathcal{K}_{P,\delta} = \left\{ \Delta \in \mathbb{R}^m : \Delta_i > \delta,\ \sum_{i=1}^{m} \Delta_i = P \right\}. \tag{1}$$

It is an $(m-1)$-dimensional open simplex. The finite-dimensional parameter manifold is

$$\widetilde{\mathcal{Q}} = (\mathbb{R}^d)^n \times \mathcal{K}_{P,\delta}. \tag{2}$$

The spline-evaluation map

$$\mathcal{E} : \widetilde{\mathcal{Q}} \longrightarrow C^r(S^1, \mathbb{R}^d), \qquad \mathcal{E}(q,\Delta)(u) = C_{q,\Delta}(u), \tag{3}$$

selects the regular stratum

$$\mathcal{Q}_{\mathrm{imm}} = \left\{ \xi \in \widetilde{\mathcal{Q}} : \partial_u \mathcal{E}(\xi)(u) \neq 0 \text{ for all } u \right\}, \tag{4}$$

and the topology-preserving stratum

$$\mathcal{Q}_{\mathrm{emb}} = \left\{ \xi \in \mathcal{Q}_{\mathrm{imm}} : \mathcal{E}(\xi) \text{ is injective on } S^1 \right\}. \tag{5}$$

The intended admissible configuration manifold $\mathcal{Q}_{\mathrm{adm}}$ is an open subset of one of these strata together with any interval, curvature, or collision margins required by the application. The present code does not enforce $\mathcal{Q}_{\mathrm{emb}}$ as a hard invariant and can therefore leave the embedded-curve stratum.

A parameterized curve and its geometric shape are not the same object. If orientation-preserving reparameterizations are immaterial, the associated shape space is represented formally by

$$\mathcal{S}_{\mathrm{par}} = \mathrm{Emb}(S^1, \mathbb{R}^d) / \mathrm{Diff}^+(S^1). \tag{6}$$

If translation and rotation are also irrelevant, one further quotients by $\mathrm{SE}(d)$. Shape spaces of immersed or embedded curves and their Riemannian metrics are generally infinite-dimensional quotient manifolds or orbifolds rather than Euclidean vector spaces [7], [8]. The finite B-spline family supplies a finite-dimensional charted model whose image lies inside this larger curve space. Its true target is therefore an orbit

$$[C_\star] = \left\{ g \circ C_\star \circ \varphi : g \in \mathcal{G}_{\mathrm{pose}},\ \varphi \in \mathrm{Diff}^+(S^1) \right\}, \tag{7}$$

where $\mathcal{G}_{\mathrm{pose}}$ is either the identity or a selected subgroup of $\mathrm{SE}(d)$. In a fixed finite B-spline representation, arbitrary reparameterizations need not remain exactly inside the same basis family, so the quotient is implemented approximately through re-sampling, arc-length measures, cyclic alignment, or optimization over a restricted reparameterization family.

2.2 Coordinate chart for a closed B-spline with latent knot intervals

Let

$$q = (q_1, \ldots, q_n), \qquad q_i \in \mathbb{R}^d,$$

denote the ambient coordinates of the control vertices. For the fixed-knot baseline, the curve is evaluated with a uniform periodic cubic basis. For the adaptive evaluator, the code stores one logit per periodic interval,

$$\gamma = (\gamma_1, \ldots, \gamma_m) \in \mathbb{R}^m, \qquad m = n,$$

and, in the safe fixed-period chart, maps it to $\mathcal{K}_{P,\delta}$ by

$$\Delta_i = \delta + (P - m\delta)\,\mathrm{softmax}(\gamma)_i. \tag{8}$$

Because $\Delta(\gamma + c\mathbf{1}) = \Delta(\gamma)$, the logits contain a one-dimensional gauge. An intrinsic chart may impose $\mathbf{1}^\top \gamma = 0$, work directly with $m-1$ coordinates, or project both γ and p_γ onto $\mathbf{1}^\perp$. The current implementation stores all m logits and therefore carries a redundant coordinate direction. The legacy chart $\Delta_i = \mathrm{softplus}(\gamma_i)$ remains available, but it does not impose fixed period and is not the manifold chart used for the main formulation.

The cumulative intervals define a periodic knot vector, and the curve is evaluated by a differentiable Cox–de Boor recursion. A small interval approaches a lower-dimensional boundary stratum associated with knot coincidence. For a degree-p spline, the continuity at a knot of multiplicity r is, under standard nondegeneracy assumptions, C^{p-r} [9], [10]. Thus exact repeated-knot activation belongs naturally to a stratified extension of $\widetilde{\mathcal{Q}}$; within the strict interval floor, the computation remains on one smooth stratum.

The discrete observation map is

$$\mathcal{O}_N(\xi) = (C_\xi(u_1), \ldots, C_\xi(u_N)) \in (\mathbb{R}^d)^N, \tag{9}$$

where $\xi = (q, \Delta)$ or its logit-chart representation (q, γ). The target point cloud is $Y = \{y_b\}_{b=1}^M \subset \mathbb{R}^d$, and the associated empirical measures are

$$\mu_\xi = \frac{1}{N}\sum_{a=1}^{N} \delta_{C_\xi(u_a)}, \qquad \nu_Y = \frac{1}{M}\sum_{b=1}^{M} \delta_{y_b}. \tag{10}$$

A matching functional $\mathcal{D}(\mu, \nu_Y)$ defines the pullback potential

$$V_Y(\xi) = \mathcal{D}\big(\mu_\xi, \nu_Y\big). \tag{11}$$

Exact reparameterization invariance requires the observation itself to be invariant, for example by using normalized arc-length measure and a sufficiently accurate quadrature. Fixed parameter samples generally break that invariance at the discrete level.

2.3 Tangent motion, pullback covectors, and a configuration metric

For a path $\xi(t) = (q(t), \Delta(t)) \in \mathcal{Q}_{\mathrm{adm}}$, the represented curve path is $C_t = \mathcal{E}(\xi(t))$. Its physical velocity is not the coordinate vector $(\dot{q}, \dot{\Delta})$ itself but its pushforward

$$\dot{C}_t(u) = d\mathcal{E}_{\xi(t)}[\dot{\xi}(t)](u) = \sum_i N_i^p(u;\Delta)\dot{q}_i + \sum_j \frac{\partial C_{q,\Delta}(u)}{\partial \Delta_j}\dot{\Delta}_j. \tag{12}$$

In the logit chart, $\dot{\Delta} = D\chi_\gamma \dot{\gamma}$, where χ denotes the fixed-period interval map. Thus $\dot{\xi} \in T_\xi \mathcal{Q}$ and $\dot{C}_t \in T_{C_t} \operatorname{Imm}(S^1, \mathbb{R}^d)$ live in different tangent spaces connected by $d\mathcal{E}_\xi$.

A Riemannian metric g_ξ is needed to identify covectors and tangent velocities. One finite-dimensional metric induced by sampled curve motion is

$$g_\xi(v, w) = \sum_{a=1}^{N} \omega_a \langle d\mathcal{E}_\xi[v](u_a), d\mathcal{E}_\xi[w](u_a)\rangle + \beta_\Delta \langle v_\Delta, w_\Delta\rangle_{\mathcal{K}}, \tag{13}$$

possibly augmented by first- or higher-order Sobolev terms along the curve. In local coordinates this metric has a mass matrix $M(\xi)$. Automatic differentiation of V_Y returns the coordinate components of the differential $dV_Y(\xi) \in T_\xi^* \mathcal{Q}$; the metric gradient is

$$\operatorname{grad}_g V_Y = \sharp_g dV_Y \equiv M(\xi)^{-1} \nabla_{\mathrm{coord}} V_Y. \tag{14}$$

The current code uses a constant diagonal M, so its update is a valid chart-based preconditioned dynamics but not an intrinsic shape-space gradient. The port-Hamiltonian formulation below is naturally expressed in terms of the covector dH and does not require calling the raw coordinate derivative an intrinsic gradient.

2.4 Information-geometric hierarchy: task, local measurement, and global pose

The observational space for a spline might differ from is configurational space. For a planar spline, each control point satisfies $q_i \in \mathbb{R}^2$. The global pose group SE(2) acts *collectively* on the complete control polygon,

$$(R, a) \cdot (q_1, \ldots, q_n, \Delta) = (Rq_1 + a, \ldots, Rq_n + a, \Delta), \tag{15}$$

while the knot intervals are unchanged. Likewise, the physical knot variable is the vector

$$\Delta \in \mathcal{K}_{P,\delta} = \left\{\Delta \in \mathbb{R}^m : \Delta_i > \delta,\ \mathbf{1}^\top \Delta = P\right\}, \tag{16}$$

which is an $(m-1)$-dimensional open simplex. The stored logit vector satisfies $\gamma \in \mathbb{R}^m / \operatorname{span}\{\mathbf{1}\}$ because a common shift does not change Δ. Based on this particular realization in 2D space, three information-geometric objects then enter the deformation problem at distinct levels.

Task-level discrepancy. A divergence $\mathcal{D}(\mu_\xi, \nu_Y)$ compares the observed spline and target and induces the scalar potential $V_Y = \mathcal{D} \circ \mathcal{O}_N$. It determines what constitutes task progress. It does not, by itself, define a metric on all configuration directions or separate pose, shape, and parameterization.

Configuration-level measurement Fisher matrix. Let a deterministic geometric or safety measurement be

$$e(\xi; Y) = h\big(\mathcal{O}_N(\xi), \xi; Y\big) \in \mathbb{R}^{m_e}, \qquad L(\xi) = \frac{\partial e}{\partial \xi} = \begin{bmatrix} L_q & L_\gamma \end{bmatrix}. \tag{17}$$

For a noisy readout $\widetilde{e} = e + \varepsilon_e$ with covariance Σ_e, or for a deterministic Gauss–Newton weighting $W \succeq 0$, the local pullback metric is

$$\mathcal{I}_{\mathrm{meas}}(\xi) = L(\xi)^\top W L(\xi) = \begin{bmatrix} L_q^\top W L_q & L_q^\top W L_\gamma \\ L_\gamma^\top W L_q & L_\gamma^\top W L_\gamma \end{bmatrix}. \tag{18}$$

The diagonal blocks measure sensitivity to spatial control-point motion and knot-parameterization motion. The cross block quantifies local confounding: a large $L_q^\top W L_\gamma$ means that the monitored residual can be changed in similar ways by moving control points or redistributing knot intervals. The knot-logit gauge must be projected out before interpreting rank or conditioning.

If the residual is rigid-motion invariant, differentiation of equation (15) gives

$$L(\xi)B_G(\xi) = 0, \qquad \mathcal{I}_{\text{meas}}(\xi)B_G(\xi) = 0, \tag{19}$$

where B_G collects the two translation generators and one rotation generator. Thus the local safety metric is naturally horizontal: it measures intrinsic shape and knot changes after global pose directions are removed.

Lie-algebra-level Souriau–Fisher matrix. On the phase space $T^*\mathcal{Q}$, the cotangent-lifted SE(2) action has momentum map

$$\mathbf{J}_{\text{spl}}(q, \gamma, p_q, p_\gamma) = \left(\sum_{i=1}^{n} q_i \times p_{q_i}, \sum_{i=1}^{n} p_{q_i} \right) \in \mathfrak{se}(2)^*. \tag{20}$$

The knot momentum does not appear because SE(2) acts trivially on γ. For an ensemble of phase-space states and a Souriau Gibbs parameter $\beta \in \mathfrak{se}(2)$, the corresponding Fisher metric is

$$\mathcal{I}_{\text{S}}(\beta) = \text{Cov}_{\pi_\beta}[\mathbf{J}_{\text{spl}}]\,. \tag{21}$$

For planar pose this is a 3×3 covariance matrix on the Lie-algebra parameter space. It describes statistical conditioning of total angular and linear momentum across an ensemble. It does not measure local pointwise shape sensitivity, and it supplies no knot block.

Consequently, the useful meaning of "one level above" is that the Souriau–Fisher construction is built from the momentum map of a Hamiltonian group action on phase space, whereas the measurement FIM is built from the Jacobian of a configuration observation. After choosing a centroid/orientation gauge and projecting the knot-logit gauge, a local pose–shape–parameterization split motivates the hybrid information structure

$$\mathcal{G}_{\text{info}} = \mathcal{I}_{\text{S},pose} \oplus \mathcal{I}_{\text{meas,shape/knot}}. \tag{22}$$

This direct-sum notation is local and requires an explicit horizontal/vertical decomposition. The two blocks should not be added as matrices in the redundant (q, γ) chart because they act on different spaces and represent different statistical questions.

2.5 Continuity adaptivity versus topological change

Within $\mathcal{K}_{P,\delta}$ with $\delta > 0$, the knot coordinates remain on a smooth simplex stratum and can only concentrate parameter resolution. Sending selected intervals to zero approaches strata with repeated knots and reduced spatial continuity. The union of these fixed-multiplicity strata is better viewed as a stratified configuration space than as one smooth manifold. This continuity adaptivity should not be confused with changing the topology of the represented image. A single embedded closed spline remains one closed component unless the representation is augmented with splitting, merging, re-meshing, multiple components, or an implicit interface. Conversely, leaving $\mathcal{Q}_{\text{emb}}$ through a self-intersection is a loss of injectivity, not a valid topology-changing operation. Circle-to-crescent and oval-to-star experiments therefore test sharp-feature formation, parameter redistribution, and preservation of the embedded stratum, not a change of genus or connectedness.

2.6 Finite-horizon optimal-control problem on $T^*\mathcal{Q}$

Let $\xi \in \mathcal{Q}_{\text{adm}}$ denote the generalized spline configuration and let $p \in T^*_\xi\mathcal{Q}_{\text{adm}}$ be its conjugate momentum. The phase-space state is

$$z = (\xi, p) \in T^*\mathcal{Q}_{\text{adm}}. \tag{23}$$

In the chart used by the code, this is stored as $x = (q, \gamma, p_q, p_\gamma)$, with the caveat that the m-logit fixed-period representation contains one gauge direction. The intrinsic phase-space dimension is $2(nd + m - 1)$ on the smooth fixed-period stratum, while the stored tensor has dimension $2(nd + m)$ unless gauge projection is applied.

If a particular target configuration ξ_1 is known and representation matters, one may use a Riemannian distance $d_{\mathcal{Q}}(\xi(T), \xi_1)$. If only the geometric target shape matters, the terminal cost should instead compare the orbit $\pi_{\mathrm{sh}}(\mathcal{E}(\xi(T)))$ with $[C_\star]$. A formal quotient-space terminal discrepancy is

$$d_{\mathcal{S}}^2\big([\mathcal{E}(\xi(T))], [C_\star]\big) = \inf_{g \in \mathcal{G}_{\mathrm{pose}},\, \varphi \in \mathrm{Diff}^+(S^1)} d_{\mathrm{curve}}^2\big(\mathcal{E}(\xi(T)), g \circ C_\star \circ \varphi\big). \tag{24}$$

When only a target observation Y is available, $V_Y(\xi(T))$ is used and the goal is the terminal set $\mathcal{A}_Y^{\varepsilon}$, not a single coordinate vector.

The finite-horizon learning problem is therefore

$$\min_{\theta, \phi, K_p, z(\cdot)} \Phi_{\mathrm{term}}\big(z(T); [C_\star], Y\big) + \int_0^T \ell\big(z(t), \theta, \phi\big)\, dt, \tag{25}$$

subject to

$$\dot{z} = F_{\theta,\phi}(z; Y) \in T_z(T^*\mathcal{Q}_{\mathrm{adm}}), \qquad z(0) = z_0. \tag{26}$$

A representative intrinsic objective is

$$\begin{aligned} \Phi_{\mathrm{term}} &= \alpha_{\mathcal{S}} d_{\mathcal{S}}^2\big([\mathcal{E}(\xi(T))], [C_\star]\big) + \alpha_Y V_Y(\xi(T)) \\ &\quad + \alpha_p\, g_{\xi(T)}^{-1}(p(T), p(T)) + \alpha_\gamma B_\gamma(\xi(T)), \end{aligned} \tag{27}$$

$$\ell(z) = \alpha_V V_Y(\xi) + \alpha_H [\dot{H}_d]_+ + \alpha_v\, g_\xi(\dot{\xi}, \dot{\xi}) + \alpha_u \|u\|^2. \tag{28}$$

One need not activate both the quotient-shape term and the observation term in the same experiment. The current code implements a finite-rollout, single-chart approximation with observation-based terminal and path losses rather than solving the continuous-time manifold optimality system directly.

Coordinate convention used below

All subsequent vectors, matrices, and symbols $\nabla_x H$ are local-coordinate representations. Geometrically, $dH(z) \in T_z^*(T^*\mathcal{Q})$ is a covector, J and R are bundle maps from cotangent to tangent spaces, and G maps port inputs into tangent vectors. The matrix formulas remain valid in a chosen cotangent chart, but the learned neural maps are not automatically equivariant under a change of spline coordinates.

3 From feedback optimal control to a port-Hamiltonian policy class

3.1 What a direct feedback solution would require

Let $\mathcal{V}(t, x; Y)$ be the value function of equation (25). Under the classical assumptions needed for a differentiable dynamic-programming solution, $\mathcal{V}$ satisfies the Hamilton–Jacobi–Bellman equation [11]

$$-\partial_t \mathcal{V} = \min_u \left\{ \ell_0(x; Y) + \frac{1}{2} u^\top Q_u u + \nabla_x \mathcal{V}^\top \big(f_0 + Gu\big) \right\}, \qquad \mathcal{V}(T, x; Y) = \Phi(x; s_1, Y). \tag{29}$$

Because the control cost is strictly convex, the formal minimizing feedback is

$$u^*(t, x; Y) = -Q_u(x)^{-1} G(x)^\top \nabla_x \mathcal{V}(t, x; Y), \tag{30}$$

and substitution into equation (29) gives

$$-\partial_t \mathcal{V} = \ell_0 + \nabla_x \mathcal{V}^\top f_0 - \frac{1}{2} \nabla_x \mathcal{V}^\top G Q_u^{-1} G^\top \nabla_x \mathcal{V}. \tag{31}$$

Equivalently, Pontryagin's maximum principle introduces the optimal-control Hamiltonian

$$\mathcal{H}_{\mathrm{oc}}(x, \lambda, u) = \ell_0(x; Y) + \frac{1}{2} u^\top Q_u u + \lambda^\top \big(f_0(x; Y) + G(x) u\big), \tag{32}$$

with stationarity $u^* = -Q_u^{-1} G^\top \lambda$, state dynamics, and a terminal-value costate equation [12]. Solving this system requires either a high-dimensional HJB equation or a target-dependent two-point boundary-value problem.

For the spline problem, both routes are difficult. On the fixed-period smooth stratum, the intrinsic phase-space dimension is $2(nd+m-1)$, while the current redundant logit chart stores $2(nd+m)$ scalar coordinates; the target-dependent potential contains a nonlocal spline evaluation and a finite-iteration transport solve; and a new target generally changes both the terminal condition and the feedback field. Moreover, a generic neural feedback law has no built-in power balance, so numerical errors or extrapolation can inject energy even when the training objective decreases on the sampled trajectories. These considerations motivate replacing the unrestricted feedback class by a structured, lower-risk class rather than claiming to solve the HJB equation directly.

3.2 The pH ansatz as a structured feedback restriction

Suppose the uncontrolled deformation model admits, exactly or approximately, an energy-based conservative part

$$f_{\mathrm{c}}(x;Y) = J_0(x)\nabla_x H_0(x;Y), \qquad J_0^\top = -J_0, \tag{33}$$

and suppose feedback enters through a power port $G(x)u$. A desired shaped energy $H_d(x;Y)$ and interconnection $J_\theta(x)$ cannot in general be selected arbitrarily. Energy shaping first requires an interconnection-matching control u_{es} satisfying

$$f_0(x;Y) + G(x)u_{\mathrm{es}}(x;Y) = J_\theta(x)\nabla_x H_d(x;Y). \tag{34}$$

This is the familiar matching requirement behind passivity-based energy shaping; a solution need not exist for a given actuation map G and desired H_d [6]. When the matching equation is satisfied exactly, or when its residual is represented and penalized approximately, one can add damping injection through

$$u_{\mathrm{di},\theta}(x;Y) = -K_\theta(x)G(x)^\top\nabla_x H_d(x;Y), \qquad K_\theta(x) \succeq 0. \tag{35}$$

Then

$$Gu_{\mathrm{di},\theta} = -GK_\theta G^\top \nabla_x H_d = -R_\theta \nabla_x H_d, \qquad R_\theta := GK_\theta G^\top \succeq 0, \tag{36}$$

and the matched closed loop becomes

$$\dot{x} = [J_\theta(x) - R_\theta(x)]\,\nabla_x H_d(x;Y). \tag{37}$$

Thus a pH model may be read as a *restricted feedback policy class*: J_θ learns how energy is exchanged among control-point, knot, and momentum variables, while R_θ learns or assigns the directions in which energy is dissipated. The present code parameterizes the closed-loop field directly and does not separately solve equation (34); this is another reason to call the construction a pH template rather than a completed IDA–PBC synthesis.

The connection with equation (30) is structural rather than an identity. If H_d approximated the value function $\mathcal{V}$, and if

$$R_\theta(x) \approx G(x)Q_u(x)^{-1}G(x)^\top,$$

then the dissipative component of equation (37) would have the same gradient-feedback form as the HJB minimizer. Exact optimality would additionally require the matching condition

$$f_0(x;Y) + G(x)u^*(t,x;Y) = [J_\theta(x) - R_\theta(x)]\,\nabla_x H_d(t,x;Y) \tag{38}$$

and zero HJB residual. Neither condition is enforced by the present implementation. In the current model, H_d is a physical/search energy, not a trained value function, and the pH network is optimized directly through rollout losses. Consequently, the method should be described as a passivity-structured approximation to the feedback-control problem, not as an exact HJB or PMP solver.

An optional value-consistency extension would introduce the residual

$$\mathcal{E}_{\mathrm{HJB}} := \partial_t H_d + \ell_0 + \nabla H_d^\top f_0 - \frac{1}{2}\nabla H_d^\top G Q_u^{-1} G^\top \nabla H_d, \tag{39}$$

and penalize $|\mathcal{E}_{\mathrm{HJB}}|^2$ during training. This is not currently implemented and would require a time-dependent energy/value parameterization and a clearly specified actuation matrix G.

3.3 What is gained, and what is not

For an autonomous closed-loop system with $H_d \in C^1$, locally Lipschitz J_θ, R_θ, $J_\theta^\top = -J_\theta$, and $R_\theta \succeq 0$, every classical solution satisfies

$$\frac{d}{dt} H_d(x(t);Y) = -\nabla H_d^\top (R_\theta) \nabla H_d \leq 0. \tag{40}$$

This provides a target-independent structural certificate that is unavailable to a generic feedback network. It also yields interpretable diagnostics: conservative power, dissipated power, force norms, and the directions in which the learned model is allowed to exchange or remove energy.

However, equation (40) is a Lyapunov inequality, not an optimality theorem. Nonincrease of H_d does not imply convergence to the desired target unless H_d is lower bounded, its relevant sublevel sets are precompact, and the largest invariant subset of

$$\left\{x :\ (R_\theta)^{1/2} \nabla H_d(x) = 0\right\}$$

contains only acceptable equilibria; a LaSalle-type argument is then required [13]. The pH decomposition is also nonunique, and a generic optimal closed-loop vector field need not admit the chosen finite-dimensional pH parameterization.

The implemented learning problem is therefore

$$\min_{\theta,\phi,K_p} \mathcal{J}\big(\kappa_{\theta,\phi,K_p}; x_0, Y\big), \qquad \kappa_{\theta,\phi,K_p} \in \mathcal{K}_{\mathrm{pH}}, \tag{41}$$

where $\mathcal{K}_{\mathrm{pH}}$ is the family generated by skew interconnections, PSD dissipations, the selected energy, and the numerical integrator. Finite-rollout training approximates equation (41); it avoids solving the global HJB/PMP system, while trading exact feedback optimality for structural stability, interpretability, and reusable target-conditioned dynamics.

4 Measure-space matching potentials

4.1 Entropic optimal transport and Sinkhorn divergence

For uniform weights $a_i = 1/N$ and $b_j = 1/M$, let

$$C_{ij} = \|x_i - y_j\|^2.$$

The entropy-regularized transport cost can be written as

$$\mathsf{OT}_\varepsilon(a, b; C) = \min_{\pi \in \Pi(a,b)} \langle C, \pi \rangle + \varepsilon \, \mathrm{KL}(\pi \,\|\, a \otimes b), \tag{42}$$

where $\Pi(a, b)$ is the set of couplings with marginals a, b. The debiased Sinkhorn divergence is

$$\mathsf{S}_\varepsilon(X, Y) = \mathsf{OT}_\varepsilon(X, Y) - \frac{1}{2}\mathsf{OT}_\varepsilon(X, X) - \frac{1}{2}\mathsf{OT}_\varepsilon(Y, Y). \tag{43}$$

The self-interaction terms are essential: they alter both the scalar value and the gradient acting on the source samples.

4.2 Ambient and adversarial latent variants

The ambient potential uses equation (43) directly on points in $\mathbb{R}^d$. The optional latent model introduces a discriminator or feature map

$$D_\phi : \mathbb{R}^d \to \mathbb{R}^{d_z}$$

and computes

$$V_{\mathrm{lat}}(q, \gamma; \phi) = \mathsf{S}_\varepsilon \left(D_\phi \# \mu_{q,\gamma}, D_\phi \# \nu \right). \tag{44}$$

The discriminator is updated by gradient ascent on the divergence, while the spline dynamics minimize it. The current implementation regularizes the feature magnitude, but it does not enforce a global Lipschitz constraint. Without bounded features or an equivalent normalization, the maximization can be poorly scaled. Moreover, the envelope-theorem interpretation is only approximate because the discriminator receives a small number of alternating ascent updates rather than being solved to equilibrium at every state.

4.3 What OT does and does not guarantee

The coupling in equation (42) preserves discrete mass marginals. Entropic OT generally gives a soft many-to-many coupling, not a bijection. It also does not enforce that the parametric curve remains injective. Avoiding self-intersection requires either an explicit barrier, a repulsive energy, an orientation/Jacobian constraint, or an ambient diffeomorphic flow such as LDDMM [14]. Therefore self-intersection count is an evaluation metric and a possible regularizer, not a theorem inherited automatically from Sinkhorn divergence.

Algorithm 1: feedback-gated pH deformation of one B-spline toward a target point cloud

Inputs: initial control polygon q_0, initial interval logits γ_0, target point cloud $Y = \{y_b\}_{b=1}^{N_Y}$, spline degree p, parameter samples $u_{1:N}$, outer stages S, inner lengths L_s, nominal state step h, stage schedule $\mathcal{S}$ from equation (72), Sinkhorn iteration count K, total knot period P_0, interval floor $\delta > 0$, learned parameters $(\theta, \widehat{k}_{\text{ctrl}})$, optional discriminator D_ϕ, active-set budgets (B_q, B_γ), gate refresh period, full-refresh period, backtracking factor $\beta \in (0, 1)$, sufficient-decrease constant $c_H \in (0, 1)$, numerical tolerance $\eta_{\text{num}} \geq 0$, maximum backtracking trials N_{bt}, and mode `train` or `eval`.
Initialize: set $p_{q,0} = 0$, $p_{\gamma,0} = 0$, $x_{0,0} = (q_0, \gamma_0, p_{q,0}, p_{\gamma,0})$, and $P_{0,0} = I$. During an optional spatial warm-up, set the knot entries of the configuration mask to zero so that the spline first translates and bends before its parameterization changes.

For each outer stage $s = 0, \ldots, S - 1$**:**
Step 0. Freeze the current environment. Set $(\varepsilon_s, \sigma_s, \lambda_s) = \mathcal{S}(\tau_s)$ using equation (72) and construct one frozen target Y_{σ_s} using equation (76). Hold $(\varepsilon_s, \sigma_s, \lambda_s, Y_{\sigma_s})$ fixed for all $k = 0, \ldots, L_s - 1$. When the curriculum is disabled, $S = 1$ and this step simply reads the constant hyperparameters from the experiment configuration.

For each inner state step $k = 0, \ldots, L_s - 1$**:**

Step 1. Read the current spline state. Unpack

$$x_{s,k} = (q_{s,k}, \gamma_{s,k}, p_{q,s,k}, p_{\gamma,s,k}).$$

The first two blocks are generalized coordinates; the last two are their conjugate momenta. Only in the canonical mechanical limit do the momentum-to-rate identities reduce to $\dot{q} = p_q / M_q$ and $\dot{\gamma} = p_\gamma / M_\gamma$.

Step 2. Construct admissible knot intervals and sample the current curve. The implemented safe chart is

$$\omega_{s,k,j} = \frac{\exp(\gamma_{s,k,j})}{\sum_r \exp(\gamma_{s,k,r})}, \qquad \Delta_{s,k,j} = \delta + (P_0 - m\delta)\omega_{s,k,j},$$

and evaluate

$$X_{s,k} = \{C_{q_{s,k}, \Delta_{s,k}}(u_a)\}_{a=1}^{N}.$$

Store the nonzero degree-p basis functions and, when needed, the sensitivities of each sample with respect to nearby control points and intervals. The fixed-period chart is present in the repository and is used by the safe-knot configurations. Selecting the legacy `raw_softplus` chart reproduces the earlier global-scale degeneracy discussed after equation (47).

Step 3. Compute the finite-iteration Sinkhorn comparison. Set $Z_X = X_{s,k}$ and $Z_Y = Y_{\sigma_s}$ in ambient mode, or $Z_X = D_\phi(X_{s,k})$ and $Z_Y = D_\phi(Y_{\sigma_s})$ in latent mode. For each pair $AB \in \{XY, XX, YY\}$, form the squared-distance matrix C^{AB}, initialize $f^{AB,(0)} = 0$ and $g^{AB,(0)} = 0$, and perform exactly K paired log-domain updates from equations (83) and (84). Thus $K = 1$ means one f update followed by one g update for each of the three cross/self problems.
Reconstruct the finite-iterate coupling

$$\pi_{ij}^{AB,(K)} = a_i b_j \exp\left(\frac{f_i^{AB,(K)} + g_j^{AB,(K)} - C_{ij}^{AB}}{\varepsilon_s} \right).$$

Because the last operation is a g update, its column marginal equals b up to roundoff and $\sum_{ij} \pi_{ij}^{AB,(K)} = 1$. Hence the correction term in equation (88) vanishes for the returned iterate, and equation (85) is the finite-iterate dual value under the code's convention. Form the corresponding debiased divergence

$$\widehat{\mathsf{S}}_{\varepsilon_s,K}(Z_X, Z_Y) = \widehat{\mathsf{OT}}^{XY}_{\varepsilon_s,K} - \tfrac{1}{2}\widehat{\mathsf{OT}}^{XX}_{\varepsilon_s,K} - \tfrac{1}{2}\widehat{\mathsf{OT}}^{YY}_{\varepsilon_s,K}.$$

There is no universal iteration count: K is an explicitly reported inner-solver hyperparameter. The repository configurations use $K = 60$ by default, while the observed stable reproduction uses $K = 1$. Increasing K changes the finite-iterate force field and should be accompanied by source-row marginal-residual logging and retuning of h, ε_s, and dissipation.
Define $\omega_{s,k,j} = \Delta_{s,k,j}/P_0$ and use

$$V_{s,k} = \widehat{\mathsf{S}}_{\varepsilon_s,K}(Z_X, Z_Y) + \lambda_s\, \mathcal{R}_{\text{int}}(\omega_{s,k}).$$

A bounded concentration surrogate such as $\mathcal{R}_{\text{int}}(\omega) = -\sum_j \omega_j \log(\omega_j + \delta)$ may be used after warm-up; it is a continuity-selection surrogate, not a proof of a repeated-knot event.

Step 4. **Convert mismatch into Hamiltonian forces.** Form

$$H_{d,s,k} = T(p_{q,s,k}, p_{\gamma,s,k}) + V_{s,k}$$

and compute $g_{s,k} = \nabla_x H_{d,s,k}$. The task forces are

$$F_{q_i,s,k} = -\nabla_{q_i} V_{s,k}, \qquad F_{\gamma_j,s,k} = -\partial_{\gamma_j} V_{s,k}.$$

The current algorithm uses the plant-only model in equation (55); no auxiliary controller state is advanced. A joint dynamic controller would instead require the block system in equation (59).

Step 5. **Choose which local spline variables should move.** At a gate-refresh step, compute the force/residual scores in equations (71) and (80), select $\mathcal{A}_q^{s,k}$ and $\mathcal{A}_\gamma^{s,k}$, and close the sets under degree-p B-spline support. The configuration selector is a matrix, not a scalar:

$$M_{s,k} = \operatorname{diag}\Big(m^q_{s,k,1} I_d, \ldots, m^q_{s,k,n} I_d, m^\gamma_{s,k,1}, \ldots, m^\gamma_{s,k,m}\Big) \in \mathbb{R}^{(nd+m)\times(nd+m)},$$

with binary entries for a hard active set or entries in $[0, 1]$ for a soft gate. Construct

$$P_{s,k} = \operatorname{diag}(M_{s,k}, M_{s,k}) \in \mathbb{R}^{2(nd+m)\times 2(nd+m)}.$$

Selecting q_i selects all d coordinates of q_i and all of p_{q_i}; selecting γ_j selects p_{γ_j} as well. Periodic full refreshes set $P_{s,k} = I$ for one step. The present repository implements the restricted scalar gate

$$P_{s,k} = \operatorname{diag}(I_q,\, g_k I_\gamma,\, I_{p_q},\, g_k I_{p_\gamma}), \qquad g_k \in [0, 1],$$

including knot warm-up with $g_k = 0$. Support-aware per-control-point and per-knot feedback selection remains the proposed extension in section 6.3.

Step 6. **Construct the learned pH matrices by parameterization.** Let the PHT output unconstrained matrices $\widehat{A}_\theta(x_{s,k})$ and $\widehat{B}_\theta(x_{s,k})$. Use the implemented residual parameterization

$$J_\theta(x_{s,k}) = J_0 + \rho\left(\widehat{A}_\theta - \widehat{A}_\theta^\top\right),$$
$$R_\theta(x_{s,k}) = \rho \widehat{B}_\theta \widehat{B}_\theta^\top + \epsilon_R I,$$

or the diagonal option in equation (53). Set

$$R_{\text{ctrl}} = \operatorname{diag}\Big(\operatorname{softplus}(\widehat{k}_{\text{ctrl}})\Big), \qquad R_{d,s,k} = R_\theta(x_{s,k}) + R_{\text{ctrl}}.$$

This reproduces the legacy full-state assigned damping. The implemented safe structural mode instead sets

$$J_\theta = J_0, \qquad R_\theta = \operatorname{diag}(0_q, 0_\gamma, D_q, D_\gamma),$$

with each diagonal entry of D_q and D_γ bounded between configured positive limits, and it disables the separate controller damping. These constructions imply $J_\theta^\top = -J_\theta$ and $R_{d,s,k} \succeq 0$ by construction; see equations (51) to (53) and (57).

Step 7. Advance the selected state and distinguish the two implemented solvers. Form

$$A_{s,k} = P_{s,k} \left[J_\theta(x_{s,k}) - R_{d,s,k}\right] P_{s,k}, \qquad \dot{x}_{s,k} = A_{s,k} g_{s,k}.$$

The learned PHT engine uses the unsafeguarded full-state forward-Euler step

$$x_{s,k+1} = x_{s,k} + h\dot{x}_{s,k}.$$

The separate canonical baseline does *not* use this update: it first advances momentum with the potential force and explicit damping multiplier $(1-hr/M)$ and then advances configuration using the new momentum. Neither implementation currently performs backtracking, substepping, or an acceptance test. Consequently, neither code path inherits the continuous dissipation inequality for arbitrary h. When a discrete certificate is required, add energy acceptance or replace the proposal by the discrete-gradient step equations (63) and (64).

Step 8. Diagnose the deformation. Record the curve, ambient and latent target errors, H_d, V, T, the finite-K marginal residual, measured and predicted power, force and speed norms, accepted step size, interval heatmap, active indices, curvature, area, length, self-intersections, and step acceptance. These diagnostics determine whether the next gate remains local, activates knot variables, or returns to $P_{s,k} = I$.

Step 9. Update learned components and advance the outer stage. In `train` mode, assemble the graph-connected rollout loss in equation (65) and update $(\theta, \widehat{k}_{\mathrm{ctrl}})$ and any differentiable gate parameters. Do not use detached diagnostic copies of H or V as training terms. In adversarial mode, detach the current spline state and update D_ϕ by constrained ascent at its prescribed frequency. In `eval` mode, keep all learned parameters fixed. Set $x_{s+1,0} = x_{s,L_s}$ when continuing the same deformation task. Change $(\varepsilon_s, \sigma_s, \lambda_s)$ only at the next outer-stage boundary and report the resulting energy jump separately from within-stage dissipation.

How Algorithm 1 relates to the current code

The current learned-PHT training path corresponds to $S = 1$, constant $(\varepsilon, \sigma, \lambda)$, a finite-iterate Sinkhorn dual value, and unsafeguarded full-state explicit Euler. The fixed-period interval chart, mechanically constrained PHT, and paired global (γ, p_γ) gate are integrated and exposed through the safe configurations. The separate curriculum path changes $(\varepsilon, \sigma, \lambda)$ across stages but does not jointly train the PHT. Source-row marginal-residual logging is available in diagnostic scripts rather than the core potential; support-aware local gating, energy backtracking, and the discrete-gradient alternative remain unintegrated.

5 Port-Hamiltonian deformation model

Section 3 motivates the pH dynamics as a restricted feedback class rather than an exact solution of the unrestricted HJB/PMP problem. We now propose constructions of pH deformation model with parametrization and refer to Algorithm 1 for computational details.

5.1 Deformation Hamiltonian

Let

$$z = (q, \gamma), \qquad p = (p_q, p_\gamma), \qquad x = (z, p).$$

The deformation Hamiltonian used by the current solver is

$$H_d(x; Y) = T(p_q, p_\gamma) + V(q, \gamma; Y), \tag{45}$$

with kinetic energy

$$T(p_q, p_\gamma) = \frac{1}{2M_q}\|p_q\|^2 + \frac{1}{2M_\gamma}\|p_\gamma\|^2. \tag{46}$$

A larger M_γ slows knot dynamics relative to control-point motion. The potential used by the current solver is

$$V(q, \gamma; Y) = V_{\mathrm{match}}(q, \gamma; Y) + \lambda_{\mathrm{int}}\, \mathcal{R}_{\mathrm{int}}(\gamma), \tag{47}$$

where V_{match} can be Sinkhorn, latent Sinkhorn, sliced Wasserstein, deterministic projected Wasserstein, KL, Bregman, MMD, or a nearest-neighbor baseline. Under the legacy `raw_softplus` chart the code retains $\mathcal{R}_{\mathrm{int}}(\gamma) = \sum_i \mathrm{softplus}(\gamma_i)$ for compatibility. Under the implemented fixed-period chart it instead penalizes squared relative log spacing around the uniform interval.

Audit of the current interval penalty

Because the adaptive evaluator scales the sampled parameter range by the total interval sum, uniformly shrinking all legacy intervals can leave the sampled geometry nearly unchanged while decreasing $\sum_i \Delta_i$. The legacy term therefore contains a global collapse direction and is not a well-posed local stacking penalty. The repository now provides the fixed-period repair

$$\Delta_i = \delta + (P_0 - m\delta)\,\mathrm{softmax}(\gamma)_i,$$

which removes the common-shift/global-scale direction and enforces a hard floor. Its current regularizer is

$$\mathcal{R}_{\mathrm{spacing}}(\gamma) = \frac{1}{m}\sum_i \left[\log\left(\frac{\Delta_i}{P_0/m}\right)\right]^2.$$

This stabilizes interval identification but does not itself promote exact repeated knots. Reconstructed knot multiplicities or interval-floor events, rather than only $\gamma_i < \tau$, should be used to certify continuity changes.

5.2 Placement of information-geometric terms in the pH model

The three information-geometric objects enter different terms of the deformation dynamics.

1. The target divergence remains in V_{match} and therefore contributes the nominal task covector dV_{match}.
2. The measurement Fisher matrix acts naturally through a measurement-aligned dissipative port or, separately, as an inverse-Fisher mobility for the horizontal shape/knot directions.
3. The Souriau–Fisher matrix acts on an explicitly separated global-pose or reduced momentum-statistics block, or is retained as a symmetry-conditioning diagnostic when no reduced pose state is introduced.

Let $v_\xi = \partial H_d/\partial p$ denote the configuration velocity and let L be the measurement Jacobian in equation (17). Closing the measurement output $y_e = Lv_\xi$ with $u_e = -D_e y_e$ produces the momentum force

$$f_{\mathrm{meas}} = -L^\top D_e L v_\xi. \tag{48}$$

When $D_e = \alpha W$, the induced damping matrix is $\alpha \mathcal{I}_{\mathrm{meas}}$. This realization preserves the interpretation of controller power,

$$y_e^\top u_e = -y_e^\top D_e y_e \le 0, \tag{49}$$

and requires only Jacobian–vector and vector–Jacobian products. It is preferable to inserting an opaque dense Fisher matrix into every state block.

Inverse-Fisher mobility is a different operation. On a horizontal basis Z_{hor}, a preconditioned task direction is obtained by solving

$$\left(Z_{\mathrm{hor}}^\top \mathcal{I}_{\mathrm{meas}} Z_{\mathrm{hor}} + \lambda M_0\right) v = -Z_{\mathrm{hor}}^\top dV_{\mathrm{match}}, \qquad \delta\xi = Z_{\mathrm{hor}} v, \tag{50}$$

without explicitly forming a matrix inverse. This changes conditioning and acceleration; it should not be reported as passive controller damping.

For the Souriau block, one first computes an ensemble covariance of equation (20). If the state has been explicitly decomposed into a pose coordinate $g \in \mathrm{SE}(2)$ and horizontal shape/knot coordinates, a regularized 3×3 Souriau metric may define a pose-space mobility or statistical trust region. In the unreduced control-point chart, the safer use is diagnostic: monitor its rank and condition number, test simultaneous spline–target equivariance, and check the forced momentum balance. A fixed target generally breaks pose-momentum conservation even when the task family is equivariant.

5.3 Learned plant interconnection and dissipation

The learned plant matrices are parameterized as

$$J_\theta(x) = J_0 + \rho\left(A_\theta(x) - A_\theta(x)^\top\right), \tag{51}$$

$$R_\theta(x) = \rho B_\theta(x) B_\theta(x)^\top + \epsilon_R I, \tag{52}$$

or, in the more stable diagonal option,

$$R_\theta(x) = \operatorname{diag}(\rho \ \mathrm{softplus}(b_\theta(x)) + \epsilon_R)\,. \tag{53}$$

Thus $J_\theta^\top = -J_\theta$ and $R_\theta \succeq 0$ by construction. The residual scale ρ is initialized small, so training begins near the canonical solver rather than an arbitrary learned flow.

The constrained ablation additionally implements the mechanical structure

$$J_\theta(x) = J_0, \qquad R_\theta(x) = \operatorname{diag}(0_q, 0_\gamma, D_q(x), D_\gamma(x))\,, \tag{54}$$

where each diagonal coefficient is obtained from a sigmoid and is bounded between user-specified lower and upper limits. This mode removes direct configuration damping and suppresses the learned skew residual; it is the repository's current "safe field" option, not merely a future recommendation.

The present network embeds the entire flattened state as a *single token* before the Transformer encoder. With sequence length one, self-attention does not perform relational attention across control points or knots; the model behaves primarily as a state-conditioned dense network with Transformer feed-forward and normalization blocks. A genuinely geometric Transformer should tokenize control points, knot intervals, and momentum pairs separately, with positional or periodic encodings and support-aware attention.

5.4 Effective assigned damping and the inert shadow controller state

The effective implemented plant dynamics add an algebraic, positive-semidefinite damping matrix to the phase-space vector field:

$$\dot{x} = [J_\theta(x) - R_\theta(x) - R_{\mathrm{ctrl}}]\,\nabla_x H_d(x;Y), \qquad R_{\mathrm{ctrl}} \succeq 0. \tag{55}$$

Here R_{ctrl} is *assigned closed-loop damping*; it is not the dissipation matrix of a separate controller subsystem. A mechanically standard choice damps only the momentum block,

$$R_{\mathrm{ctrl}}^{\mathrm{mom}} = \begin{bmatrix} 0 & 0 \\ 0 & D_{\mathrm{ctrl}} \end{bmatrix}, \qquad D_{\mathrm{ctrl}} = \operatorname{diag}\Big(\mathrm{softplus}(\widehat{d}_{\mathrm{ctrl}})\Big) \succeq 0, \tag{56}$$

with the block ordering $x = (z, p)$. The optional legacy controller branch instead permits a full-state diagonal parameterization

$$R_{\mathrm{ctrl}}^{\mathrm{full}} = \operatorname{diag}\Big(\mathrm{softplus}(\widehat{k}_{\mathrm{ctrl}})\Big) \succeq 0. \tag{57}$$

The full-state form is mathematically dissipative but is better interpreted as state-space damping or preconditioning, because it acts directly on configuration-gradient directions as well as momentum directions.

For the autonomous continuous-time model in equation (55),

$$\frac{dH_d}{dt} = -\nabla H_d^\top\,(R_\theta + R_{\mathrm{ctrl}})\,\nabla H_d \le 0. \tag{58}$$

This is the power balance relevant to the effective model. The engine can additionally allocate a tensor x_c and define $H_c = \frac{1}{2}\sum_i K_{p,i}(x_{c,i} - x_i)^2$. However, training initializes $x_c = x$ and updates $x_c^+ = x_c + h\dot{x}$ with exactly the same $\dot{x}$ used for $x^+ = x + h\dot{x}$. Therefore $x_c - x$ is invariant and $H_c \equiv 0$ for the standard initialization. The only nontrivial effect of K_p is the added diagonal matrix R_{ctrl}. Safe-structure configurations disable this controller branch entirely.

5.5 Optional dynamic controller: joint block-matrix model

A genuine dynamic controller is a different construction. Let x_c be a controller state with Hamiltonian $H_c(x_c)$, interconnection $J_c(x_c) = -J_c(x_c)^\top$, dissipation $R_c(x_c) \succeq 0$, and plant–controller port map $G_c(x, x_c)$. The joint state must then satisfy the block pH system

$$\begin{bmatrix} \dot{x} \\ \dot{x}_c \end{bmatrix} = \begin{bmatrix} J_\theta(x) - R_\theta(x) - R_{\text{ctrl}} & -G_c(x, x_c) \\ G_c(x, x_c)^\top & J_c(x_c) - R_c(x_c) \end{bmatrix} \begin{bmatrix} \nabla_x H_d(x; Y) \\ \nabla_{x_c} H_c(x_c) \end{bmatrix}. \tag{59}$$

The off-diagonal blocks are skew transposes and therefore exchange power without creating it. For the total energy

$$H_{\text{tot}}(x, x_c; Y) = H_d(x; Y) + H_c(x_c), \tag{60}$$

one obtains

$$\frac{dH_{\text{tot}}}{dt} = -\nabla_x H_d^\top (R_\theta + R_{\text{ctrl}}) \nabla_x H_d - \nabla_{x_c} H_c^\top R_c \nabla_{x_c} H_c \leq 0. \tag{61}$$

In this joint model, R_c is the internal dissipation of the controller subsystem and must not be identified with the plant-level matrix R_{ctrl} in equation (55). A static feedback reduction of equation (59) may produce an effective plant damping matrix, but that reduction must be derived; one should not simultaneously retain x_c, add $H_c(x, x_c)$ to the plant energy, and insert the same gain again as R_{ctrl}.

A nontrivial Energy–Casimir design further requires a Casimir relation $\mathcal{C}(x, x_c) = \text{const}$ whose invariance follows from matching conditions on $J_\theta, J_c, G_c, R_\theta$, and R_c. Merely setting $x_c(0) = x(0)$ and advancing x_c with the same derivative as x makes $x_c - x$ identically zero, but it does not realize a meaningful dynamic controller or energy-shaping law. The current repository therefore reduces numerically to equation (55); the allocated shadow state does not realize equation (59).

5.6 Discrete state update

The learned engine currently advances the plant-only state by explicit Euler:

$$x_{k+1} = x_k + h \left[J_\theta(x_k) - R_\theta(x_k) - R_{\text{ctrl}}\right] \nabla H_d(x_k; Y). \tag{62}$$

This is the implemented update, but it is not by itself a symplectic or dissipativity-preserving integrator. The continuous inequality equation (58) does not guarantee $H_{d,k+1} \leq H_{d,k}$ for arbitrary h. The minimum practical safeguard is explicit Euler with backtracking or substepping and an admissibility test, as stated in Step 7 of Algorithm 1.

A stronger alternative is a discrete-gradient pH step

$$\frac{x_{k+1} - x_k}{h} = \left[\overline{J}_k - \overline{R}_k\right] \overline{\nabla} H_d(x_k, x_{k+1}), \tag{63}$$

where $\overline{J}_k^\top = -\overline{J}_k$, $\overline{R}_k \succeq 0$, and the discrete gradient satisfies $\overline{\nabla} H_d(x_k, x_{k+1})^\top (x_{k+1} - x_k) = H_d(x_{k+1}) - H_d(x_k)$. Then

$$H_d(x_{k+1}) - H_d(x_k) = -h \overline{\nabla} H_d^\top \overline{R}_k \overline{\nabla} H_d \leq 0. \tag{64}$$

This provides a genuine discrete energy certificate at the cost of an implicit nonlinear solve [15].

6 Learning and two-time-scale schemes

6.1 Finite-rollout training

For a rollout of L steps, the intended loss is

$$\begin{aligned} \mathcal{L} = V(x_L) + \alpha_{\text{path}} \frac{1}{L} \sum_{k=1}^{L} V(x_k) + \alpha_H \frac{1}{L-1} \sum_{k=1}^{L-1} [H_{k+1} - H_k]_+ \\ + \alpha_p \left(\|p_q^L\|^2 + \|p_\gamma^L\|^2\right) + \alpha_\gamma B_\gamma(\gamma^L). \end{aligned} \tag{65}$$

The optimizer updates PHT parameters and controller damping. When the adversarial potential is active, a separate optimizer periodically performs discriminator ascent.

The present implementation uses a first-order approximation to differentiable physics. The Hamiltonian gradient ∇H is computed with `create_graph=False` unless the caller explicitly enables force differentiation. Consequently, gradients reach the PHT through J_θ and R_θ and through the terminal potential, but the rollout does not differentiate through the Hessian of the Sinkhorn potential at every step. This reduces memory and often improves robustness, but it should be described as a stop-gradient-force approximation.

The diagnostic tensors are detached by default at the engine API, but the training script explicitly calls `forward_step(..., detach_diagnostics=False)`. Hence the stored H_k and V_k are graph connected and the path-potential and positive energy-increment terms contribute gradients through earlier rollout states. The first path term is evaluated at the fixed initial state and therefore has no direct PHT-parameter gradient; later terms do. The remaining approximation is that ∇H itself is treated as a stop-gradient vector unless `--differentiate-forces` is enabled. The energy-increment sequence also uses pre-step Hamiltonians and does not include the final post-step increment $H(x_L) - H(x_{L-1})$.

6.2 Adversarial alternating update

The optional latent scheme alternates:

1. a deformation/PHT minimization step for V_{lat};
2. one or more discriminator ascent steps for $V_{\mathrm{lat}} - \lambda_D \|D_\phi(\cdot)\|^2$.

This is an empirical min–max procedure, not an exact evaluation of $\max_\phi V_{\mathrm{lat}}$. Stable use generally requires feature normalization, gradient or spectral penalties, controlled ascent frequency, and evaluation against the ambient-space metric to detect latent distortion.

6.3 Feedback-gated subset updates

With the default gate $g_k = 1$, the learned rollout updates every component of q, γ, p_q, and p_γ at every state step. The implemented warm-up can freeze the complete (γ, p_γ) pair, but it cannot yet select individual local supports. A computationally cheaper and often better conditioned extension is to update only those local spline degrees of freedom for which the present deformation exhibits large force, matching residual, curvature mismatch, or stagnation. This produces a hybrid active-set controller whose selector is recomputed from feedback. In this subsection, s denotes the outer environment stage and k the inner state step, consistently with Algorithm 1.

Let the configuration vector be $z = (q, \gamma)$, let $d_z = nd + m$, and introduce a diagonal configuration selector

$$M_{s,k} = \mathrm{diag}\Big(m^q_{s,k,1} I_d, \ldots, m^q_{s,k,n} I_d, m^\gamma_{s,k,1}, \ldots, m^\gamma_{s,k,m}\Big), \qquad 0 \le m_{s,k,i} \le 1. \tag{66}$$

The same selector must be applied to each coordinate and its conjugate momentum:

$$P_{s,k} = \mathrm{diag}(M_{s,k}, M_{s,k}). \tag{67}$$

A hard active set uses binary entries; a differentiable gate uses entries in $[0, 1]$. The structure-preserving masked flow is

$$\dot{x}_{s,k} = P_{s,k}\big[J_\theta(x_{s,k}) - R_d(x_{s,k})\big] P_{s,k}\, g_{s,k}, \qquad R_d = R_\theta + R_{\mathrm{ctrl}}, \qquad g_{s,k} = \nabla_x H_d(x_{s,k}). \tag{68}$$

Because $P_{s,k} = P_{s,k}^\top$, the masked matrices

$$J^{\mathrm{act}}_{s,k} = P_{s,k} J_\theta P_{s,k}, \qquad R^{\mathrm{act}}_{s,k} = P_{s,k} R_d P_{s,k} \tag{69}$$

remain skew-symmetric and positive semidefinite, respectively. For a frozen curriculum stage,

$$\dot{H}_d = -(P_{s,k} g_{s,k})^\top R_d (P_{s,k} g_{s,k}) \le 0. \tag{70}$$

This property generally fails for a one-sided mask such as $P_{s,k}(J - R_d) g_{s,k}$, since $g_{s,k}^\top P_{s,k} J g_{s,k}$ need not vanish. It is also important to activate a complete canonical pair: selecting q_i activates all spatial components of q_i and p_{q_i}; selecting γ_j activates both γ_j and p_{γ_j}.

The cheapest feedback score is the normalized generalized-force magnitude,

$$s^{q}_{s,k,i} = \frac{\|F_{q_i,s,k}\|}{\mathrm{median}_r \|F_{q_r,s,k}\| + \delta}, \qquad s^{\gamma}_{s,k,j} = \frac{|F_{\gamma_j,s,k}|}{\mathrm{median}_r |F_{\gamma_r,s,k}| + \delta}. \tag{71}$$

When a transport plan or barycentric projection is available, these scores can be augmented by local matching residuals pulled back through the B-spline basis and knot sensitivities. Hard top-B selection is appropriate for evaluation, while smooth sigmoid or sparsemax gates are preferable during end-to-end training. Hysteresis, a minimum dwell time, local spline-support closure, and periodic full-state refresh steps prevent gate chattering and permanently frozen regions. The complete deformation rollout and active-set construction are stated next in section 6.5; the numbered operations correspond directly to the blocks in figure 3.

6.4 Stagewise environment schedule and index convention

The environment parameters must not be changed inside a state step if the energy balance is to have a fixed meaning. We therefore use an outer stage index $s = 0, \ldots, S-1$ and an inner deformation-step index $k = 0, \ldots, L_s - 1$. Set

$$\tau_s = \frac{s}{\max(S-1, 1)}, \qquad (\varepsilon_s, \sigma_s, \lambda_s) = \mathcal{S}(\tau_s), \tag{72}$$

where $\mathcal{S}$ is a prescribed schedule. The three values are held constant throughout the inner rollout of stage s and are changed only at a stage boundary. If no curriculum is used, take $S = 1$ and keep all three values constant for the complete rollout.

One reproducible cooling schedule is

$$\varepsilon_s = \varepsilon_{\min} + (\varepsilon_{\max} - \varepsilon_{\min})(1 - \tau_s)^{r_\varepsilon}, \tag{73}$$

$$\sigma_s = \sigma_{\max}(1 - \tau_s)^{r_\sigma}, \tag{74}$$

$$\lambda_s = \lambda_{\max}\, \chi_s\, \tau_s^{r_\lambda}, \tag{75}$$

where $\chi_s \in \{0, 1\}$ activates interval adaptation only after a spatial warm-up or a matching-potential plateau. The direction of the λ_s schedule depends on the sign and meaning of the interval regularizer. In this manuscript, λ_s multiplies a *concentration-promoting* continuity-selection surrogate under a fixed total knot period, so it is zero during spatial warm-up and may increase afterward. If λ_s instead multiplies a barrier that resists interval collapse, its schedule must be interpreted oppositely. The unnormalized prototype term $\lambda \sum_j \mathrm{softplus}(\gamma_j)$ has the scale degeneracy identified in the audit after equation (47) and should not be used to justify a general scheduling rule.

For a point-cloud target, a stagewise Gaussian smoothing can be represented by

$$Y_{\sigma_s} = \{y_b + \sigma_s \xi_{s,b}\}_{b=1}^{N_Y}, \qquad \xi_{s,b} \sim \mathcal{N}(0, I), \tag{76}$$

with the sampled perturbations frozen during the entire inner rollout. Resampling Y_{σ_s} at every state step would make the potential explicitly stochastic/time-dependent and would invalidate a direct within-stage dissipation comparison unless the injected supply is modeled separately.

6.5 Task-transparent spline deformation algorithm

The deformation task can now be read directly from the variables in the model. The control points q_i determine where the spline lies in physical space; the interval variables γ_j determine how parameter space is distributed along the closed curve; and the conjugate momenta (p_q, p_γ) store phase-space momentum. They are not automatically deformation rates: for the canonical kinetic block, $\dot{q} = p_q/M_q$ and $\dot{\gamma} = p_\gamma/M_\gamma$, while learned residual interconnection/dissipation can modify these relations. At each state step, the sampled source curve is compared with the target point cloud, the matching potential generates spatial and interval forces, the feedback rule decides which local spline supports need to move, and the pH operator converts those forces into a dissipative phase-space update.

6.5.1 Feedback rule for local control-point and knot activation

The selector in Step 5 is a computational policy, not a second geometric model. Let $\mathcal{A}_q^{s,k-1}$ and $\mathcal{A}_\gamma^{s,k-1}$ denote the previously active sets within stage s. A low-overhead implementation uses the normalized force scores in equation (71). If a coupling $\pi_{s,k}$ is reconstructed, define a sample-wise barycentric mismatch

$$d_{s,k,a} = \left\| \frac{\sum_b \pi_{s,k,ab} y_b}{\sum_b \pi_{s,k,ab} + \delta} - x_{s,k,a} \right\|, \tag{77}$$

and pull it back to the spline degrees of freedom:

$$r^q_{s,k,i} = \frac{\sum_a |N_i(u_a)| d_{s,k,a}}{\sum_a |N_i(u_a)| + \delta}, \tag{78}$$

$$r^\gamma_{s,k,j} = \frac{1}{N} \sum_a \left\| \frac{\partial C_{q_{s,k},\gamma_{s,k}}(u_a)}{\partial \gamma_j} \right\| d_{s,k,a}. \tag{79}$$

After robust normalization, use

$$\widetilde{s}^q_{s,k,i} = \alpha_F \|\widehat{F_{q_i,s,k}}\| + \alpha_R \widehat{r}^q_{s,k,i}, \qquad \widetilde{s}^\gamma_{s,k,j} = \beta_F |\widehat{F_{\gamma_j,s,k}}| + \beta_R \widehat{r}^\gamma_{s,k,j}. \tag{80}$$

Control points with the largest scores are activated immediately. Knot variables are preferably activated only after the matching potential has plateaued over a window W,

$$\text{plateau}_{s,k} = \mathbf{1}\left[\frac{V_{s,k-W} - V_{s,k}}{W \max(V_{s,k-W}, \delta)} < \eta_V \right], \tag{81}$$

so that continuity adaptation is used only after ordinary spatial motion is insufficient. Top-B selection, on/off hysteresis, minimum dwell time, local support closure, and periodic full refreshes prevent chattering and permanently frozen regions. A soft learned gate $w_\psi(x) \in [0,1]^{n+m}$ may replace the hard selector during training, provided the same gate is replicated on the coordinate and momentum blocks and appears on both sides of $(J - R_d)$.

6.5.2 Recommended task schedule

For circle-to-crescent and oval-to-star deformation, the default schedule is:

1. begin with all control points active and all knot variables frozen;
2. allow the control polygon to remove translation and smooth low-frequency mismatch;
3. when the Sinkhorn potential stagnates, activate only the largest knot-force regions and the control-point neighborhoods in their degree-p support;
4. freeze a neighborhood only when both its local matching residual and force fall below their off-thresholds;
5. use periodic full-state refreshes and energy-based step acceptance throughout the rollout.

This schedule makes each subset update traceable to a visible part of the current spline rather than to an opaque global mask.

6.6 Thermodynamic curriculum

The repository contains a separate slow-time curriculum module with

$$\varepsilon = \varepsilon(\tau), \qquad \sigma = \sigma(\tau), \qquad \lambda = \lambda(\tau), \qquad \tau \in [0,1]. \tag{82}$$

At each curriculum stage, the target point cloud is perturbed/smoothed with scale $\sigma(\tau)$, the Sinkhorn temperature is cooled, and the interval penalty is changed. An inner fixed pH solver runs until a kinetic-energy threshold or a step limit is reached.

This curriculum is *implemented as a separate experimental path*, but it is not yet integrated with the learned PHT training loop, adversarial discriminator training, or Fisher-preconditioned natural-gradient updates. Natural-gradient model refinement remains a proposal [16].

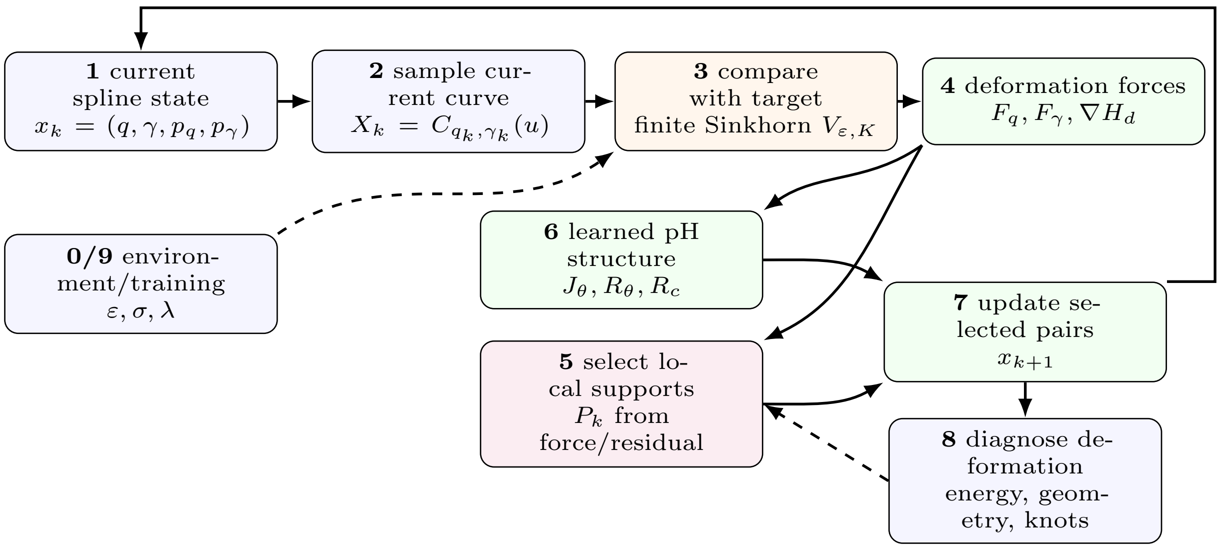


Figure 3: Task-level computational pipeline corresponding directly to Algorithm 1 in section 6.5. The source spline is sampled, compared with the target, converted into control-point and knot forces, optionally restricted to local spline supports, and advanced by the learned pH operator. Setting $P_k = I$ recovers the default full-state mode; the implemented scalar knot gate is a restricted block-diagonal instance of P_k. Applying the selector on both sides of the pH operator and to complete coordinate–momentum pairs preserves skew interconnection and PSD dissipation pointwise.

7 Finite-iteration Sinkhorn computation

7.1 Log-domain recursion used by the code

With $f^{(0)} = 0$ and $g^{(0)} = 0$, the implementation repeats K alternating updates:

$$f_i^{(k+1)} = -\varepsilon \log \sum_j b_j \exp\left(\frac{-C_{ij} + g_j^{(k)}}{\varepsilon}\right), \tag{83}$$

$$g_j^{(k+1)} = -\varepsilon \log \sum_i a_i \exp\left(\frac{-C_{ij} + f_i^{(k+1)}}{\varepsilon}\right). \tag{84}$$

It then returns

$$\widehat{\mathsf{OT}}_{\varepsilon,K}(X,Y) = a^\top f^{(K)} + b^\top g^{(K)}, \tag{85}$$

and forms

$$\mathsf{S}_{\varepsilon,K}(X,Y) = \widehat{\mathsf{OT}}_{\varepsilon,K}(X,Y) - \tfrac{1}{2}\widehat{\mathsf{OT}}_{\varepsilon,K}(X,X) - \tfrac{1}{2}\widehat{\mathsf{OT}}_{\varepsilon,K}(Y,Y). \tag{86}$$

Each potential evaluation therefore performs three Sinkhorn loops. The arithmetic cost is

$$O\big(K(NM + N^2 + M^2)\big), \tag{87}$$

and the differentiated computation graph grows linearly with K.

7.2 Finite-iterate dual interpretation

For arbitrary potentials (f, g), the KL-regularized dual objective associated with the code's convention is

$$\mathcal{D}_\varepsilon(f,g) = a^\top f + b^\top g - \varepsilon \sum_{ij} a_i b_j \left[\exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon}\right) - 1\right]. \tag{88}$$

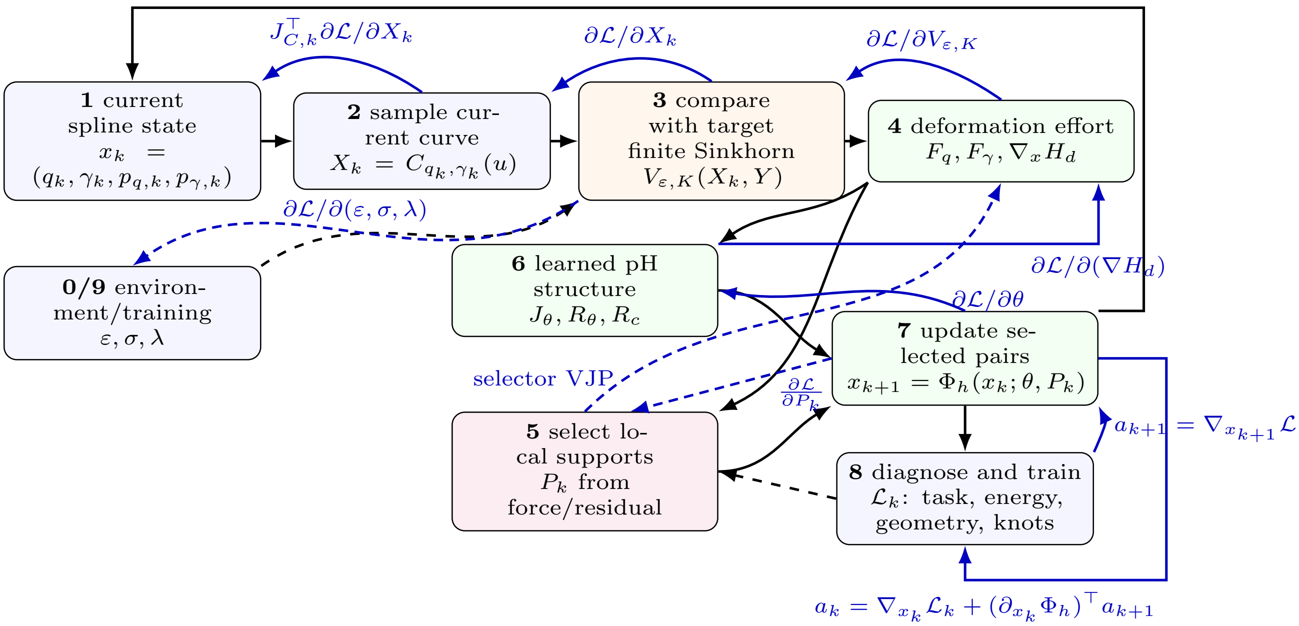


Figure 4: Task-level forward and reverse computational pipeline without the controller. **Blue reverse arrows:** solid: active end-to-end gradient; dashed: optional gradient such as hyperparameter sensitivities. Black arrows show the forward rollout. The sampled spline is compared with the target, the discrepancy is pulled back to control-point and knot efforts, the learned port-Hamiltonian operator constructs the deformation vector field, and the selected state pairs are advanced by the numerical integrator. Blue arrows show reverse-mode differentiation from the rollout loss through the update map, learned pH structure, Hamiltonian effort, finite Sinkhorn computation, and B-spline sampling map. The recurrent adjoint satisfies $a_k = \nabla_{x_k}\mathcal{L}_k + (\partial_{x_k}\Phi_h)^\top a_{k+1}$. Dashed blue paths are active only when the local selector, knot gate, or potential hyperparameters are differentiable. Setting $P_k = I$ recovers the default full-state mode. Applying the selector to complete coordinate–momentum pairs and on both sides of the pH operator preserves skew interconnection and positive-semidefinite dissipation pointwise.

Define

$$\pi_{ij}^{(K)} = a_i b_j \exp\left(\frac{f_i^{(K)} + g_j^{(K)} - C_{ij}}{\varepsilon}\right).$$

Every loop iteration ends with equation (84); therefore $\sum_i \pi_{ij}^{(K)} = b_j$ up to floating-point error and $\sum_{ij} \pi_{ij}^{(K)} = 1$. The exponential correction in equation (88) consequently vanishes, so the code does not omit that term after a complete sweep. What remains unconverged is generally the source marginal, measured by

$$r_{\mathrm{row}}^{(K)} = \left\|\pi^{(K)}\mathbf{1} - a\right\|_1.$$

Thus $K = 1$ and $K = 60$ are finite iterates of the same inner dual problem, but the unrolled algorithm produces different scalar approximations and different derivatives until the marginal residual is small.

7.3 Why one iteration can be more stable

For $K = 1$,

$$f_i^{(1)} = -\varepsilon \log \sum_j b_j e^{-C_{ij}/\varepsilon}, \tag{89}$$

which is a soft minimum of source-to-target distances. The resulting force resembles a smoothed nearest-neighbor or soft-Chamfer attraction and only weakly enforces global marginal balance. Additional iterations increasingly couple all source and target samples through the dual variables. This yields a better transport approximation but can produce:

1. stronger global competition for target mass;

2. larger force Jacobians near assignment changes, especially for small ε;
3. stronger interaction with the XX debiasing term when source samples cluster;
4. a larger effective stiffness in the learned explicit-Euler update equation (62);
5. deeper and potentially less well-conditioned differentiation through repeated log-sum-exp operations.

Therefore the observation "one iteration is more stable" does not show that one iteration is a more accurate Sinkhorn solver. It shows that the current time step, damping, temperature, sample density, and inner-iteration level are better matched to the one-step surrogate.

7.4 Dimensionless stability view

Let L_K be a local Lipschitz estimate of the learned vector field using K Sinkhorn iterations. Explicit Euler is reliable only when the dimensionless product

$$\chi_K = hL_K \tag{90}$$

is sufficiently small for the local spectrum. Increasing K can increase L_K even when the scalar potential changes modestly. A useful controller-aware indicator is

$$\chi_K^{\text{eff}} = h \left\| (J_\theta - R_\theta - R_c)\, \nabla^2 H_{d,K} \right\|_2 . \tag{91}$$

The dashboard's two-dimensional probe-point Hessian provides a local approximation to this quantity for selected control points.

8 Evaluation and validation protocol

8.1 Metrics

A case should be evaluated along four axes.

Matching. Report the configured Sinkhorn potential, ambient-space Chamfer distance, bidirectional Hausdorff approximation, and final target residual. Latent Sinkhorn alone is insufficient because the feature map may distort geometric scale.

pH structure and stability. Report

$$e_J = \frac{\|J + J^\top\|_F}{\|J\|_F + \delta}, \qquad \lambda_{\min}(R_d), \qquad \operatorname{tr}(R_d), \tag{92}$$

$$P_J = \nabla H^\top J \nabla H, \qquad P_R = \nabla H^\top R_d \nabla H, \tag{93}$$

$$e_{\text{power}} = \left| \frac{H_{k+1} - H_k}{h} - (P_J - P_R) \right| . \tag{94}$$

A valid structure should have $e_J \approx 0$, $\lambda_{\min}(R_d) \geq 0$, and $P_J \approx 0$. The measured discrete energy change may differ from the continuous prediction when h is too large. For canonical-runner artifacts, recompute H_d from each stored state before evaluating this residual, because the current logged H_d series is shifted relative to the stored post-step T and V.

Geometry. Report curve length, enclosed area for simple closed curves, curvature statistics, self-intersection count, and control-polygon movement. OT matching does not replace these checks.

Information geometry and symmetry. Report the normalized generator residual $\|LB_G\|/(\|L\|\|B_G\| + \delta)$, the corresponding null residual $\|\mathcal{I}_{\text{meas}} B_G\|$, the spectrum and condition number of the gauge- and pose-projected measurement metric, and a normalized control-point/knot confounding score based on $L_q^\top W L_\gamma$. For an ensemble, report the eigenvalues and condition number of the 3×3 Souriau–Fisher covariance. Test equivariance by transforming both the initial spline and target by the same random SE(2) element. With a fixed target, report the discrete momentum balance against the applied target/controller wrench rather than claiming momentum conservation.

Interval and continuity diagnostics. Report normalized interval heatmaps, minimum/maximum interval, interval ratio, reconstructed knot multiplicities, and continuity order at candidate corners. A raw count of $\gamma_i < -4$ is only a heuristic indicator.

8.2 Case-by-case experiments

The current 2D cases are useful in increasing order of difficulty:

1. **Circle to crescent:** tests concavity and tip formation.
2. **Oval to five-point star:** tests multiple alternating convex and concave corners.
3. **Smooth target baseline:** isolates damping and pH convergence without continuity reduction.

For each case, compare fixed knots, adaptive knots with the canonical solver, learned diagonal-R PHT, learned full-R PHT, no-controller damping, and optional adversarial latent Sinkhorn. The manuscript should report measured outcomes rather than the expected values contained in an experimental plan.

The 3D sphere-to-cylinder and torus-to-hexagonal-nut examples remain forward-looking. The repository currently supports a uniform closed tensor-product surface, but it does not yet support independent adaptive knot vectors (γ_u, γ_v) or non-toroidal boundary conditions required for a faithful sphere/cylinder construction.

8.3 Sinkhorn iteration audit

At a fixed saved state, sweep

$$K \in \{1, 2, 5, 10, 20, 60\}$$

and record:

1. $\mathsf{S}_{\varepsilon,K}$ and the full finite dual equation (88);
2. row and column marginal residuals of the reconstructed coupling;
3. $\|\nabla_q V_K\|$, $\|\nabla_\gamma V_K\|$, and cosine similarity with the $K = 60$ force;
4. largest eigenvalue of a probe-point Hessian or a finite-difference Jacobian estimate;
5. one-step ΔH for a grid of state step sizes h;
6. wall time and peak memory.

Then perform two separate comparisons:

- *matched time step:* hold h fixed to expose stiffness;
- *matched stability:* reduce h for larger K until the same energy acceptance criterion is satisfied.

This separates transport accuracy from integrator stability.

8.4 Reporting checklist

For each deformation case, report the source and target construction, number of control points, spline degree, sample counts, masses, state step, Sinkhorn temperature and iteration count, interval penalty, damping, PHT architecture, and residual scale. State whether R is full-state, diagonal full-state, or bounded momentum-only; whether the skew residual is enabled; whether force differentiation is enabled; whether latent adversarial training is active; and whether Algorithm 1 uses the global knot gate or a support-aware subset update. Report ambient and latent matching metrics, residual kinetic energy, self-intersections, certified knot multiplicities, active-set size and switches, energy/power plots, and the step-wise force-field dashboard. Ablations should vary K, ε, h, damping, adaptive knots, and the feedback selector under identical accepted physical time.

9 Recommended implementation corrections

1. **Use convergence diagnostics for finite Sinkhorn.** Keep the finite-iterate dual semantics, but log the source-row marginal residual, use a residual tolerance or a declared fixed K, and separate inner-solver accuracy from state-step stability.
2. **Decouple transport and state solvers.** Adapt the pH step size independently through energy acceptance, line search, or substepping; do not use a smaller K as an implicit state-integrator stabilizer without reporting that approximation.

3. **Separate and safeguard the two integrators.** Retain distinct names for the canonical momentum-first step and the learned full-state Euler step. Add post-step energy recomputation and energy acceptance; use discrete gradients or another dissipative geometric method when a discrete certificate is required.
4. **Use the implemented fixed-period chart as the default.** Migrate learned-knot experiments from `raw_softplus`, retain the hard floor and fixed period, and add a continuity-targeted regularizer if actual knot concentration is desired. Certify physical intervals rather than raw logits.
5. **Treat the controller branch as assigned damping until independent dynamics are added.** Remove the inert x_c copy or provide a genuine block interconnection and nontrivial controller Hamiltonian before claiming Energy–Casimir shaping.
6. **Preserve graph-connected path losses and control Hessian cost explicitly.** The current trainer already keeps H, V connected; document the stop-gradient-force approximation and enable full force differentiation selectively, by checkpointing, or by truncated backpropagation.
7. **Tokenize the geometric state.** Represent each control point and knot interval as a token to give attention a geometric meaning and reduce the need for dense $d \times d$ outputs.
8. **Constrain the adversary.** Add spectral normalization, gradient penalties, or bounded feature normalization; evaluate ambient and latent metrics together.
9. **Add self-contact and regularity terms.** Use explicit self-intersection barriers, bending/fairness energies, and orientation checks when these properties are required.
10. **Separate the information-geometric blocks in code.** Implement the target divergence, local measurement FIM, and Souriau–Fisher pose covariance as three named interfaces. Project the knot gauge and rigid generators before solving local inverse-Fisher systems, and do not concatenate the 3×3 Souriau covariance with the redundant configuration FIM without an explicit pose–shape decomposition.

10 Relation to continuum models and future extensions

LDDMM constructs diffeomorphic ambient flows and therefore provides a stronger non-folding model than point-cloud OT alone, at the cost of prohibiting topology changes and solving a larger continuum problem [14]. Level-set methods can split and merge interfaces but lose a fixed explicit parameter correspondence [1]. The present spline model occupies an intermediate position: it preserves an explicit boundary and allows adaptive continuity, but it does not yet support unrestricted topology change.

The pairwise formulation also suggests a population-level extension. Let Π_0 and Π_1 be probability distributions over spline configurations or over their observed shape measures. The desired operator is then a flow map $\mathcal{T}_t$ satisfying approximately

$$(\mathcal{T}_0)_\# \Pi_0 = \Pi_0, \qquad (\mathcal{T}_T)_\# \Pi_0 \approx \Pi_1.$$

Rather than evolve every degree of freedom for every sample, a feedback policy can decompose each trajectory into active local blocks. At time step k, masks $P_k^{(r)}$ identify control-point and knot neighborhoods with large local transport residual, force, or curvature mismatch, and the corresponding projected pH subflows update only complete coordinate–momentum pairs. The union of these local subflows defines an adaptive decomposition of the distributional transport. This construction does not by itself create or destroy connected components; genuine splitting and merging still require a multi-component or hybrid implicit–explicit representation.

A future unknown-target setting can replace the point-cloud matching potential with an image- or field-dependent energy. One example is the Mumford–Shah functional [17],

$$E(u, \Gamma) = \int_{\Omega \setminus \Gamma} |u - u_0|^2 \, dx + \mu \int_{\Omega \setminus \Gamma} |\nabla u|^2 \, dx + \nu \, \mathrm{Length}(\Gamma). \tag{95}$$

An explicit spline boundary can parameterize Γ when the number of components is fixed. Genuine junction formation, splitting, or merging would still require multiple spline components with connectivity operations or a hybrid implicit-explicit representation.

11 Conclusion

The unified framework interprets adaptive B-spline matching as a finite-horizon optimal-control problem on an augmented phase space. Sinkhorn-based and alternative point-cloud potentials provide smooth discrepancy signals, while the PHT imposes skew interconnection and PSD dissipation. The current repository implements the Euclidean spline, legacy and fixed-period knot charts, raw and latent Sinkhorn together with several alternative divergences, legacy residual and safe mechanical PHT structures, a paired global knot gate, optional assigned damping, finite-rollout training, evaluation artifacts, and a detailed diagnostic dashboard. It also contains a separate curriculum scaffold and a uniform 3D surface evaluator.

The cross-validation shows that several stronger claims must remain future work: entropic OT does not guarantee bijective deformation or absence of self-intersections; interval collapse is continuity adaptivity rather than arbitrary topology change; the current controller contributes damping but no nontrivial Casimir energy shaping; and the curriculum is not integrated with learned PHT training or natural gradients. Finally, one-step Sinkhorn stability should be interpreted as an interaction between a finite-iterate soft transport force and the state integrator, not as evidence that one iteration is the more accurate transport solution. The audit also shows that baseline energy artifacts must be realigned before they are used to support a discrete dissipation claim. These distinctions make the computational pipeline easier to audit and provide a clear roadmap from the current prototype to a defensible learning and control method.

References

[1] S. Osher and J. A. Sethian, "Fronts propagating with curvature-dependent speed: Algorithms based on hamilton–jacobi formulations," *Journal of Computational Physics*, vol. 79, no. 1, pp. 12–49, 1988.

[2] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," in *Advances in Neural Information Processing Systems*, vol. 26, 2013.

[3] G. Peyré and M. Cuturi, "Computational optimal transport," *Foundations and Trends in Machine Learning*, vol. 11, no. 5–6, pp. 355–607, 2019.

[4] J. Feydy, T. Séjourné, F.-X. Vialard, S.-i. Amari, A. Trouvé, and G. Peyré, "Interpolating between optimal transport and MMD using sinkhorn divergences," in *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics*, 2019, pp. 2681–2690.

[5] A. van der Schaft and D. Jeltsema, *Port-Hamiltonian Systems Theory: An Introductory Overview*. Now Publishers, 2014.

[6] R. Ortega, A. J. van der Schaft, B. Maschke, and G. Escobar, "Interconnection and damping assignment passivity-based control of port-controlled hamiltonian systems," *IEEE Transactions on Automatic Control*, vol. 47, no. 8, pp. 1218–1233, 2002.

[7] P. W. Michor and D. Mumford, "An overview of the riemannian metrics on spaces of curves using the hamiltonian approach," *Applied and Computational Harmonic Analysis*, vol. 23, no. 1, pp. 74–113, 2007. arXiv: math/0605009.

[8] M. Bauer, M. Bruveris, and P. W. Michor, "Overview of the geometries of shape spaces and diffeomorphism groups," *Journal of Mathematical Imaging and Vision*, vol. 50, pp. 60–97, 2014. arXiv: 1305.1150.

[9] C. de Boor, *A Practical Guide to Splines*, Revised. Springer, 2001.

[10] L. Piegl and W. Tiller, *The NURBS Book*, 2nd ed. Springer, 1997.

[11] D. P. Bertsekas, *Dynamic Programming and Optimal Control*, 4th ed. Athena Scientific, 2017, vol. 1.

[12] L. S. Pontryagin, V. G. Boltyanskii, R. V. Gamkrelidze, and E. F. Mishchenko, *The Mathematical Theory of Optimal Processes*. Interscience Publishers, 1962.

[13] H. K. Khalil, *Nonlinear Systems*, 3rd ed. Prentice Hall, 2002.

[14] M. F. Beg, M. I. Miller, A. Trouvé, and L. Younes, "Computing large deformation metric mappings via geodesic flows of diffeomorphisms," *International Journal of Computer Vision*, vol. 61, no. 2, pp. 139–157, 2005.

[15] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*, 2nd ed. Springer, 2006.

[16] S.-i. Amari, "Natural gradient works efficiently in learning," *Neural Computation*, vol. 10, no. 2, pp. 251–276, 1998.

[17] D. Mumford and J. Shah, "Optimal approximations by piecewise smooth functions and associated variational problems," *Communications on Pure and Applied Mathematics*, vol. 42, no. 5, pp. 577–685, 1989.

[18] X. Ma et al., "Towards layer-wise image vectorization," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 16 314–16 323.

[19] O. Hirschorn, A. Jevnisek, and S. Avidan, "Optimize & reduce: A top-down approach for image vectorization," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024, pp. 2148–2156.

A Divergences Between Probability Distributions

There are a number of types of divergences which can be used to measure the distance between two probability distributions. These include the following:

- Wasserstein distance
- Sinkhorn divergence
- Bregman divergence
- Kullback–Leibler (KL) divergence
- Max Entropy
- Rényi divergence

Let μ and ν denote probability measures on a measurable space $\mathcal{X}$. When densities exist with respect to a common reference measure, they are denoted by p and q, respectively. The Kullback–Leibler (KL) divergence, Bregman divergence, Wasserstein distance, and Sinkhorn divergence provide different geometric notions of discrepancy between probability distributions.

Kullback–Leibler divergence. The KL divergence from ν to μ is defined by

$$D_{\mathrm{KL}}(\mu \,\|\, \nu) = \int_{\mathcal{X}} \log\left(\frac{d\mu}{d\nu}(x)\right) d\mu(x), \tag{96}$$

whenever μ is absolutely continuous with respect to ν. In terms of densities p and q, this becomes

$$D_{\mathrm{KL}}(p \,\|\, q) = \int_{\mathcal{X}} p(x) \log\left(\frac{p(x)}{q(x)}\right) dx. \tag{97}$$

The KL divergence is generally asymmetric and does not define a metric. It is central in information geometry, where it locally induces the Fisher information metric.

Bregman divergence. Let $\phi : \Omega \to \mathbb{R}$ be a strictly convex, differentiable function on a convex domain $\Omega \subseteq \mathbb{R}^d$. The Bregman divergence generated by ϕ is

$$D_{\phi}(x \,\|\, y) = \phi(x) - \phi(y) - \langle \nabla\phi(y), x - y \rangle . \tag{98}$$

Bregman divergences measure the error between $\phi(x)$ and the first-order Taylor approximation of ϕ about y. They are generally asymmetric and need not satisfy the triangle inequality. The KL divergence is a special case of a Bregman divergence. For example, on the probability simplex, taking the negative entropy potential

$$\phi(p) = \sum_{i=1}^{n} p_i \log p_i \tag{99}$$

gives

$$D_{\phi}(p \,\|\, q) = \sum_{i=1}^{n} p_i \log\left(\frac{p_i}{q_i}\right) = D_{\mathrm{KL}}(p \,\|\, q). \tag{100}$$

Thus, KL divergence can be viewed as the entropy-induced Bregman divergence. In the code we use a quadratic function ϕ to distinguish it from KL.

Wasserstein distance. Let $(\mathcal{X}, d)$ be a metric space, and let $\Pi(\mu, \nu)$ denote the set of couplings between μ and ν, i.e.,

$$\Pi(\mu, \nu) = \{\pi \in \mathcal{P}(\mathcal{X} \times \mathcal{X}) : (\mathrm{proj}_1)_\# \pi = \mu,\ (\mathrm{proj}_2)_\# \pi = \nu\}\,. \tag{101}$$

The p-Wasserstein distance is

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p \, d\pi(x, y) \right)^{1/p}. \tag{102}$$

For $p = 2$, the squared 2-Wasserstein distance is often written as

$$W_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathcal{X} \times \mathcal{X}} \|x - y\|^2 \, d\pi(x, y). \tag{103}$$

Unlike KL and general Bregman divergences, Wasserstein distance accounts for the geometry of the underlying space $\mathcal{X}$ by measuring the cost of transporting mass from μ to ν.

Entropy-regularized optimal transport. The entropy-regularized optimal transport cost introduces an entropic penalty into the transport problem. Given a cost function $c(x, y)$ and regularization strength $\varepsilon > 0$, define

$$\mathrm{OT}_\varepsilon(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \left\{ \int_{\mathcal{X} \times \mathcal{X}} c(x, y) \, d\pi(x, y) + \varepsilon D_{\mathrm{KL}}(\pi \,\|\, \mu \otimes \nu) \right\}. \tag{104}$$

For discrete probability vectors $a \in \Delta_n$ and $b \in \Delta_m$ with cost matrix $C \in \mathbb{R}^{n \times m}$, this becomes

$$\mathrm{OT}_\varepsilon(a, b) = \min_{P \in U(a,b)} \left\{ \langle C, P \rangle + \varepsilon \sum_{i,j} P_{ij} \left(\log P_{ij} - 1 \right) \right\}, \tag{105}$$

where

$$U(a, b) = \left\{ P \in \mathbb{R}_+^{n \times m} : P\mathbf{1}_m = a,\ P^\top \mathbf{1}_n = b \right\}. \tag{106}$$

This regularized problem can be efficiently solved using Sinkhorn iterations.

Sinkhorn divergence. The entropy-regularized OT cost OT_ε is biased because $\mathrm{OT}_\varepsilon(\mu, \mu)$ is generally nonzero. The Sinkhorn divergence removes this entropic bias by defining

$$S_\varepsilon(\mu, \nu) = \mathrm{OT}_\varepsilon(\mu, \nu) - \frac{1}{2}\mathrm{OT}_\varepsilon(\mu, \mu) - \frac{1}{2}\mathrm{OT}_\varepsilon(\nu, \nu). \tag{107}$$

The Sinkhorn divergence interpolates between optimal transport and kernel-like divergences as the regularization parameter varies. In the small-regularization limit, it approaches the unregularized optimal transport cost:

$$\lim_{\varepsilon \to 0} S_\varepsilon(\mu, \nu) = W_c(\mu, \nu), \tag{108}$$

where W_c denotes the optimal transport cost associated with c. For larger ε, the entropy term becomes more dominant, connecting Sinkhorn divergences to information-geometric and Bregman-type structures.

Maximum-entropy bridge potential. Let $X = \{x_i\}_{i=1}^n$ denote points sampled from the current spline and let $Y = \{y_j\}_{j=1}^m$ denote the target point cloud. With the squared Euclidean cost $c_{ij} = \|x_i - y_j\|^2$ and temperature $\tau > 0$, the implemented directional Gibbs free energy is

$$\mathcal{F}_\tau(X \to Y) = \frac{1}{n} \sum_{i=1}^n \left[-\tau \log \left(\frac{1}{m} \sum_{j=1}^m \exp\Big(-\frac{c_{ij}}{\tau} \Big) \right) \right]. \tag{109}$$

For each source point, the associated Gibbs policy is

$$\pi_\tau(j \mid i) = \frac{\exp(-c_{ij}/\tau)}{\sum_{\ell=1}^m \exp(-c_{i\ell}/\tau)}. \tag{110}$$

The implementation optionally adds a target-marginal residual and removes source–source and target–target self terms. As $\tau \to 0$, the directional term approaches an average nearest-neighbor cost; increasing τ produces a smoother and more diffuse assignment. This is a closed-form one-step endpoint matching surrogate, not a Schrödinger bridge yet, because the current model contains neither a reference transition kernel nor a path-space relative-entropy constraint.

Rényi divergence and free energy. The current Rényi divergence option constructs normalized Gaussian KDE densities p and q for the current and target point clouds on the pooled support $Z = X \cup Y$. For order $\alpha > 0$, $\alpha \neq 1$, the discrete Rényi divergence is

$$D_\alpha(p \,\|\, q) = \frac{1}{\alpha - 1} \log\left(\sum_k p_k^\alpha q_k^{1-\alpha}\right). \tag{111}$$

At $\alpha = 1$, the implementation uses the KL limit

$$D_1(p \,\|\, q) = \sum_k p_k \log\left(\frac{p_k}{q_k}\right). \tag{112}$$

The default symmetric Rényi free-energy potential is

$$\mathcal{R}_{\alpha,\tau}(X, Y) = \frac{\tau}{2}\left[D_\alpha(p \,\|\, q) + D_\alpha(q \,\|\, p)\right]. \tag{113}$$

Here α selects the divergence geometry, τ scales the free-energy term relative to the remaining Hamiltonian energies, and the KDE bandwidth h determines the endpoint density model. Thus changing τ does not alter the KDE smoothing. This endpoint construction is a deterministic ablation; a full RCaI formulation must instead apply the Rényi divergence to controlled and optimality-conditioned trajectory laws.

Geometric interpretation. These quantities induce different geometries on probability distributions. KL divergence is an information-theoretic divergence associated with the Fisher information metric. More generally, Bregman divergences arise from convex potentials and induce dually flat information-geometric structures. Wasserstein distance instead defines a transport geometry by considering how probability mass moves across the underlying space. Sinkhorn divergences connect these perspectives by adding entropy, or equivalently KL/Bregman, regularization to the optimal transport problem. Thus, Sinkhorn-type discrepancies provide a computationally efficient bridge between Wasserstein geometry and entropy-based information geometry.

B Numerical Gradients and Measurement-Aligned Port Controllers

B.1 Open plant and power variables

Retain the phase-space ordering

$$z = (q, \gamma), \qquad p = (p_q, p_\gamma), \qquad x = (z, p), \tag{114}$$

and let $H_d(x; Y)$ be the nominal plant Hamiltonian in equation (45). Here x denotes local cotangent-bundle coordinates and $\nabla_x H_d$ denotes the coordinate representation of the differential dH_d. The open deformation plant is

$$\dot{x} = \big[J_\theta(x) - R_d(x)\big]\nabla_x H_d(x; Y) + G_q(x)u_q + G_\gamma(x)u_\gamma, \tag{115}$$

where

$$J_\theta^\top = -J_\theta, \qquad R_d = R_\theta + R_{\text{ctrl}} \succeq 0. \tag{116}$$

The power-conjugate outputs are

$$y_q = G_q(x)^\top \nabla_x H_d(x; Y), \qquad y_\gamma = G_\gamma(x)^\top \nabla_x H_d(x; Y). \tag{117}$$

Consequently,

$$\dot{H}_d = -\nabla H_d^\top R_d \nabla H_d + y_q^\top u_q + y_\gamma^\top u_\gamma. \tag{118}$$

The products $y_q^\top u_q$ and $y_\gamma^\top u_\gamma$ are controller power supplied to the plant. This is the distinction missing from a formulation that merely adds every violation term to V: a violation controller may inject or remove energy through an explicit port, and its power must be audited separately from plant dissipation.

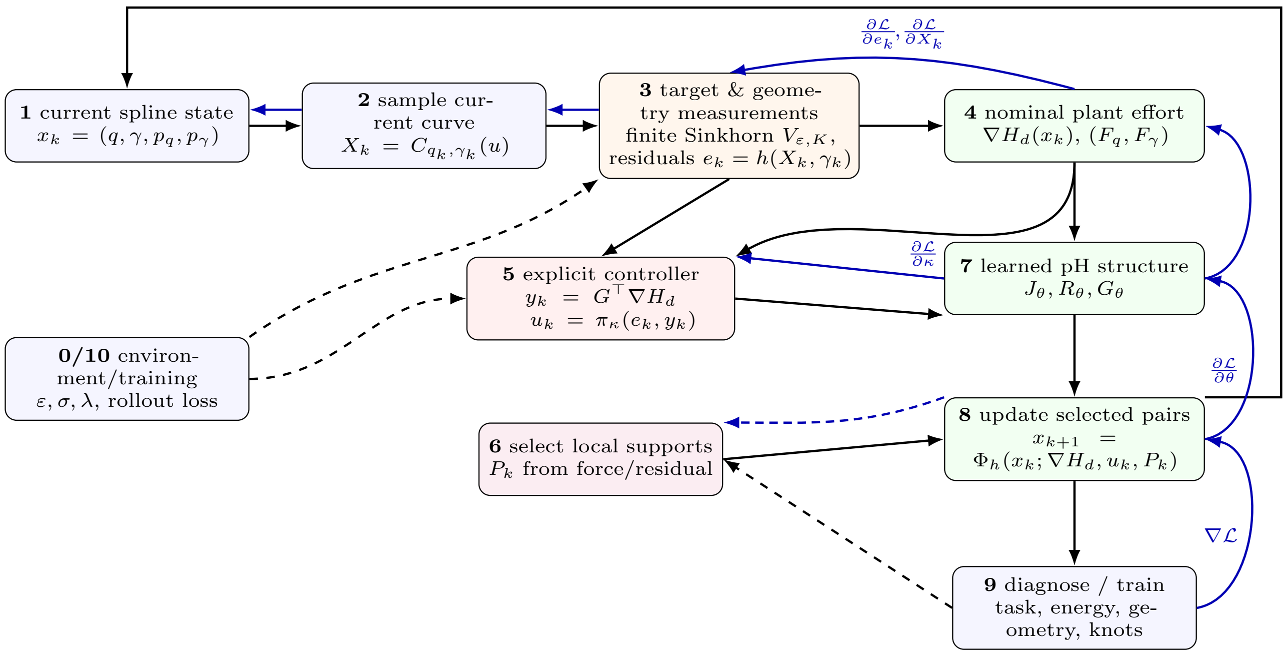


Figure 5: Task-level computational pipeline with an explicit controller port. The source spline is sampled, compared with the target, and converted into both a nominal deformation energy gradient ∇H_d and geometry-violation measurements e_k. The controller receives the measurement residuals e_k and the plant output $y_k = G^\top \nabla H_d$, then generates a correction input $u_k = \pi_\kappa(e_k, y_k)$. The learned pH block combines $(J_\theta, R_\theta, G_\theta)$, the nominal plant effort, the controller action, and the local support selector P_k to advance the spline state. Blue arrows indicate end-to-end backpropagation through the rollout loss, update map, learned pH operator, controller, measurement block, and spline sampling. Setting $P_k = I$ recovers the default full-state mode; the implemented scalar knot gate is a restricted block-diagonal instance of P_k.

B.2 Sampled B-spline measurement Jacobians

Let $u_{1:N}$ be the current parameter samples and stack the sampled curve into

$$r(q,\gamma) = \begin{bmatrix} C(u_1; q, \gamma) \\ \vdots \\ C(u_N; q, \gamma) \end{bmatrix} \in \mathbb{R}^{Nd}. \tag{119}$$

For the nonuniform B-spline

$$C(u; q, \gamma) = \sum_{i=1}^{n} N_{i,p}\big(u; \Xi(\gamma)\big) q_i, \tag{120}$$

the sample Jacobians are

$$J_q = \frac{\partial r}{\partial q}, \qquad \frac{\partial C(u_a; q, \gamma)}{\partial q_i} = N_{i,p}\big(u_a; \Xi(\gamma)\big) I_d, \tag{121}$$

$$J_\gamma = \frac{\partial r}{\partial \gamma}, \qquad \frac{\partial C(u_a; q, \gamma)}{\partial \gamma_j} = \sum_i q_i \sum_k \frac{\partial N_{i,p}(u_a; \Xi)}{\partial \Xi_k} \frac{\partial \Xi_k}{\partial \gamma_j}. \tag{122}$$

For the implemented fixed-period chart,

$$\Delta_i = \delta + (P_0 - m\delta)\omega_i, \qquad \omega_i = \frac{e^{\gamma_i}}{\sum_j e^{\gamma_j}}, \tag{123}$$

so

$$\frac{\partial \Delta_i}{\partial \gamma_j} = (P_0 - m\delta)\omega_i(\delta_{ij} - \omega_j). \tag{124}$$

The knot-vector derivative in equation (122) follows by the cumulative interval-to-knot map. This derivative is dense in the interval chart but remains locally filtered by B-spline basis support after the knot vector is assembled.

Let a vector-valued deformation measurement be

$$e = h\big(r(q,\gamma), q, \gamma, t; Y\big) \in \mathbb{R}^{m_e}. \tag{125}$$

Examples include target signed-distance samples, local orientation residuals, nonadjacent segment-separation violations, strip-Jacobian violations, interval-floor residuals, and support-density mismatch. Its configuration Jacobians are

$$L_q := \frac{\partial e}{\partial q} = h_r J_q + h_q^{\mathrm{dir}}, \tag{126}$$

$$L_\gamma := \frac{\partial e}{\partial \gamma} = h_r J_\gamma + h_\gamma^{\mathrm{dir}}, \tag{127}$$

where the direct terms account for measurements that depend explicitly on the control polygon, intervals, quadrature weights, or active support masks rather than only through sampled points. The measurement dynamics are

$$\dot{e} = L_q \dot{q} + L_\gamma \dot{\gamma} + \partial_t e. \tag{128}$$

B.3 Geometry-correction measurements and controller storage

The geometry terms discussed previously should be restored as explicit components of the measurement vector rather than appended indiscriminately to the nominal plant potential. Write

$$e_{\mathrm{geo}} = \begin{bmatrix} e_{\mathrm{nb}} \\ e_{\mathrm{xing}} \\ e_{\mathrm{orient}} \\ e_{\mathrm{sep}} \\ e_{\mathrm{strip}} \\ e_{\mathrm{knot}} \end{bmatrix}, \qquad \Psi_{\mathrm{geo}}(e_{\mathrm{geo}}) = \frac{1}{2}\sum_s e_s^\top K_s e_s, \qquad K_s \succeq 0. \tag{129}$$

The global matching discrepancy remains in H_d, while Ψ_{geo} is controller storage. Its effort

$$\lambda_{\mathrm{geo}} = \nabla_{e_{\mathrm{geo}}} \Psi_{\mathrm{geo}} \tag{130}$$

is pulled back through the measurement Jacobians and injected by the external port. This preserves the distinction between nominal task dynamics and feedback activated by violations.

Narrow-band or UDF measurement. Let ϕ_Y be a signed- or unsigned-distance field of the target and let $r_a = C(u_a; q, \gamma)$. A contour-focused residual can be written as

$$w_a = \frac{\mathrm{ReLU}(\tau_{\mathrm{nb}} - |\phi_Y(r_a)|)}{\sum_b \mathrm{ReLU}(\tau_{\mathrm{nb}} - |\phi_Y(r_b)|) + \eta}, \qquad e_{\mathrm{nb},a} = \sqrt{w_a}\, \psi_{\mathrm{nb}}(\phi_Y(r_a)), \tag{131}$$

where ψ_{nb} may be the identity, a Huber score, or another smooth robust residual. This is the B-spline analogue of the UDF-focused correction used in layer-wise vectorization [18]. If w_a is differentiated, every sample is coupled through the normalization denominator; detaching w_a yields a more local controller and should be reported as a separate numerical choice. This measurement improves target-contour localization but does not by itself prevent a self-crossing curve from lying near the target boundary.

LIVE/O&R local geometry measurements. For an ordered sample polygon, define $d_a = r_{a+1} - r_a$ and the signed local turn

$$\chi_a = d_{a-1}^{(1)} d_a^{(2)} - d_{a-1}^{(2)} d_a^{(1)}. \tag{132}$$

A smooth turn-consistency residual is

$$e_{\mathrm{orient},a} = \tanh(\beta \chi_a) - s_a^{\mathrm{ref}}, \tag{133}$$

where s_a^{ref} is obtained from the target, the previous accepted state, or a local safe orientation template. Using a fixed global convexity sign is inappropriate for a crescent because intended concavity would be penalized. For cubic spans, one may instead extract local Bézier controls (A, B, C, D) and evaluate a smooth version of the LIVE self-crossing score or the Optimize-and-Reduce orientation/intersection/angle score [18], [19]. Denote the resulting nonnegative residual by

$$e_{\mathrm{xing},s} = h_{\mathrm{LIVE/O\&R}}(A_s, B_s, C_s, D_s). \tag{134}$$

The same construction can be applied to cyclic four-sample windows when Bézier extraction is unavailable. These terms are inexpensive local anti-degeneracy measurements, not global topological certificates.

Nonadjacent segment-separation measurement. Let $\mathcal{P}$ contain nonadjacent boundary-segment pairs and let d_{ab}^{seg} be a differentiable segment-distance approximation. Define

$$e_{\mathrm{sep},ab} = \mathrm{softplus}\left(\frac{d_{\mathrm{safe}} - d_{ab}^{\mathrm{seg}}}{\tau_{\mathrm{sep}}}\right), \qquad (a, b) \in \mathcal{P}. \tag{135}$$

Unlike a local turning score, this measurement reacts to nonlocal near-collisions. Exact segment-intersection detection should still reject an unsafe proposal because a smooth distance surrogate can miss a crossing reached in one large explicit step.

Ribbon, prism, or strip quality measurement. If the boundary is thickened into a two-layer strip, let J_e be the signed area ratio of strip triangle e and Q_e a scale-free quality score. Appropriate one-sided measurements are

$$e_{J,e} = \mathrm{softplus}\left(\frac{\delta_J - J_e}{\tau_J}\right), \qquad e_{Q,e} = \mathrm{softplus}\left(\frac{Q_{\min} - Q_e}{\tau_Q}\right). \tag{136}$$

They expose local inversion and sliver risk to the q controller. Global edge–edge or triangle overlap remains a separate safety diagnostic.

Knot-safety and support-allocation measurements. For the fixed-period intervals $\Delta_j(\gamma)$, use for example

$$e_{\mathrm{floor},j} = \mathrm{softplus}\left(\frac{\delta_{\mathrm{safe}} - \Delta_j}{\tau_\Delta}\right), \tag{137}$$

$$e_{\mathrm{ratio},j} = \mathrm{softplus}\left(\frac{|\log \Delta_{j+1} - \log \Delta_j| - r_{\max}}{\tau_r}\right), \tag{138}$$

$$e_{\mathrm{alloc},j} = \frac{\Delta_j}{P_0} - \frac{\ell_j(q, \gamma)}{\sum_k \ell_k(q, \gamma) + \eta}, \tag{139}$$

where ℓ_j estimates the geometric arc length represented by interval j. The floor and ratio residuals are principally γ-port measurements, whereas allocation mismatch depends on both q and γ and may use a shared port. The safe interval chart remains the hard admissibility layer; the controller discourages concentration before the chart boundary is approached.

Role of the restored geometry terms

The UDF/narrow-band term localizes target sensing; LIVE/O&R terms provide local crossing and orientation warnings; segment separation handles nonlocal near-collision; strip Jacobian and quality measurements detect local inversion or slivers; and knot measurements protect parameterization. None alone certifies topology. The energy certificate applies to their smooth controller storage, while hard crossing, negative-Jacobian, and interval-admissibility events still require proposal rejection or backtracking.

B.4 Two equivalent force implementations, but different outputs

There are two useful ways to inject the same pulled-back correction force.

Coordinate-force port. Define identity force ports on the momenta,

$$G_q^{\text{force}} = \begin{bmatrix} 0 \\ 0 \\ I_q \\ 0 \end{bmatrix}, \qquad G_\gamma^{\text{force}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ I_\gamma \end{bmatrix}. \tag{140}$$

Then $y_q^{\text{force}} = \partial_{p_q} H_d$ and $y_\gamma^{\text{force}} = \partial_{p_\gamma} H_d$. A controller must explicitly compute the pulled-back forces

$$u_q = -L_q^\top \lambda, \qquad u_\gamma = -L_\gamma^\top \lambda. \tag{141}$$

This form is simple, but its port output is generalized velocity rather than the deformation measurement rate requested in this appendix.

Measurement-aligned port. Place the measurement Jacobian in the momentum rows:

$$G_q(x) = \begin{bmatrix} 0 \\ 0 \\ L_q(x)^\top \\ 0 \end{bmatrix}, \qquad G_\gamma(x) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ L_\gamma(x)^\top \end{bmatrix}. \tag{142}$$

For the separable kinetic energy in equation (46), define

$$v_q := \partial_{p_q} H_d = M_q^{-1} p_q, \qquad v_\gamma := \partial_{p_\gamma} H_d = M_\gamma^{-1} p_\gamma. \tag{143}$$

The conjugate outputs become

$$y_q = L_q v_q, \qquad y_\gamma = L_\gamma v_\gamma. \tag{144}$$

Under the canonical mechanical structure, where $\dot{q} = v_q$ and $\dot{\gamma} = v_\gamma$, these outputs satisfy

$$y_q + y_\gamma = \dot{e} - \partial_t e. \tag{145}$$

Thus $y = G^\top \nabla H_d$ is not an arbitrary learned readout: it is the instantaneous deformation-induced rate of the selected violation measurement.

The equality in equation (145) is exact for the safe mechanical PHT in equation (54). It need not remain exact for a learned full-state $J_\theta - R_\theta$ whose configuration rows modify $\dot{z}$ directly. In that case define the kinematic mismatch

$$\delta_e := \dot{e} - \partial_t e - (y_q + y_\gamma), \tag{146}$$

and either constrain the configuration rows to retain canonical kinematics or log δ_e as part of the certificate. A small δ_e is required before the port output is interpreted as the measured deformation rate.

B.5 Controller effort and gradient pullback

Let $\Psi(e) \geq 0$ be a controller storage associated with the violation and define

$$\lambda = \nabla_e \Psi(e). \tag{147}$$

For example,

$$\Psi(e) = \tfrac{1}{2} e^\top K_e e, \qquad \lambda = K_e e, \qquad K_e \succeq 0, \tag{148}$$

or one may use a smooth one-sided storage for inequality violations. With the measurement-aligned port, the passive proportional-damping law is

$$u = -\lambda - D_y y, \qquad D_y \succeq 0, \tag{149}$$

where $G = [G_q \; G_\gamma]$ and $y = y_q + y_\gamma$ for a shared measurement. The plant receives

$$Gu = -\begin{bmatrix} 0 \\ 0 \\ L_q^\top \lambda \\ L_\gamma^\top \lambda \end{bmatrix} - G D_y y. \tag{150}$$

Therefore the conservative part of the correction is exactly the chain-rule pullback

$$-\nabla_{(q,\gamma)}\Psi(e(q,\gamma)) = -\begin{bmatrix} L_q^\top \\ L_\gamma^\top \end{bmatrix}\lambda, \tag{151}$$

but it enters the plant through an explicit port rather than being silently absorbed into V. This distinction permits controller saturation, scheduling, activation logic, and power logging without redefining the nominal plant Hamiltonian.

For numerical implementation, neither L_q nor L_γ needs to be materialized. The correction force is a vector–Jacobian product,

$$\left(f_q^{\mathrm{ctrl}}, f_\gamma^{\mathrm{ctrl}}\right) = -\,\mathrm{VJP}_{(q,\gamma)}[e]\,\lambda, \tag{152}$$

and the measured output is a Jacobian–vector product,

$$y_q + y_\gamma = \mathrm{JVP}_{(q,\gamma)}[e]\,(v_q, v_\gamma). \tag{153}$$

A VJP/JVP implementation is preferable to constructing dense Jacobians, especially for knot sensitivities. If the rollout treats controller forces as stop-gradient quantities, only first derivatives of e are required. Differentiating through the controller force during learning introduces Hessian–vector products of e and should be enabled as a separate, explicitly reported option, analogous to `--differentiate-forces` for ∇H_d.

B.6 End-to-end gradient through the learned pH rollout

Let ϑ collect the learned PHT parameters and let κ collect controller gains, measurement temperatures, and any learned port parameters. With

$$A_\vartheta(x) = J_\vartheta(x) - R_\vartheta(x) - R_{\mathrm{ctrl}}, \qquad g(x) = \nabla_x H_d(x;Y), \tag{154}$$

the controller-augmented explicit step is

$$x_{k+1} = \Phi_h(x_k;\vartheta,\kappa) = x_k + hF_k, \tag{155}$$

with

$$F_k = A_\vartheta(x_k)g_k + G_q(x_k)u_{q,k} + G_\gamma(x_k)u_{\gamma,k}. \tag{156}$$

At step k, the differentiable computation graph is

$$x_k \longrightarrow \left(H_{d,k}, g_k, r_k, e_k, L_k, G_k, y_k, \lambda_k, u_k\right) \longrightarrow F_k \longrightarrow x_{k+1}. \tag{157}$$

The outer objective may retain the rollout loss in equation (65) and optionally add controller-aware terms,

$$\mathcal{L}_{\mathrm{tot}} = \mathcal{L} + \alpha_\Psi \frac{1}{L}\sum_{k=0}^{L-1}\Psi(e_k) + \alpha_u \frac{1}{L}\sum_{k=0}^{L-1}\|u_k\|^2 + \alpha_\delta \frac{1}{L}\sum_{k=0}^{L-1}\|\delta_{e,k}\|^2. \tag{158}$$

Adding Ψ to the outer optimization objective does not make it part of the nominal plant Hamiltonian; it remains controller storage in the closed-loop power balance.

For reverse-mode differentiation, define $a_k = \nabla_{x_k}\mathcal{L}_{\mathrm{tot}}$. If ℓ_k denotes the contribution evaluated at step k, the discrete adjoint recursion is

$$a_L = \nabla_{x_L}\ell_L, \tag{159}$$

$$a_k = \nabla_{x_k}\ell_k + \left[I + h\,\partial_x F_k\right]^\top a_{k+1}, \qquad k = L-1,\dots,0, \tag{160}$$

and the parameter gradients are

$$\nabla_\vartheta \mathcal{L}_{\mathrm{tot}} = \sum_{k=0}^{L-1}\left[\nabla_\vartheta \ell_k + h\,(\partial_\vartheta F_k)^\top a_{k+1}\right], \tag{161}$$

$$\nabla_\kappa \mathcal{L}_{\mathrm{tot}} = \sum_{k=0}^{L-1}\left[\nabla_\kappa \ell_k + h\,(\partial_\kappa F_k)^\top a_{k+1}\right]. \tag{162}$$

Automatic differentiation implements these recursions without explicitly forming $\partial_x F_k$.

The full state differential of the controlled field is

$$\mathrm{d}F = (\mathrm{d}A_\vartheta)g + A_\vartheta\,\mathrm{d}g + (\mathrm{d}G)u + G\,\mathrm{d}u, \tag{163}$$

where

$$\begin{aligned} \mathrm{d}g &= \nabla_x^2 H_d\,\mathrm{d}x, & (164)\\ \mathrm{d}e &= L\,\mathrm{d}z + \partial_t e\,\mathrm{d}t, & (165)\\ \mathrm{d}\lambda &= \nabla_e^2\Psi\,\mathrm{d}e, & (166)\\ \mathrm{d}y &= (\mathrm{d}G)^o pg + G^o p\mathrm{d}g, & (167)\\ \mathrm{d}u &= -\mathrm{d}\lambda - (\mathrm{d}D_y)y - D_y\,\mathrm{d}y. & (168) \end{aligned}$$

Thus a fully end-to-end gradient requires Hessian–vector products of both H_d and the geometry measurements. For the knot channel, these include mixed derivatives through the Cox–de Boor basis and the fixed-period softmax chart. Dense Hessians are unnecessary: VJP, JVP, and Hessian–vector-product operations are sufficient.

Three numerical differentiation modes. The current pH implementation already distinguishes whether the Hamiltonian force is graph connected. The port extension should expose an analogous controller option.

1. **First-order or stop-gradient-force mode.** Compute g_k with `create_graph=False` and detach the geometry pullback $L_k^\top\lambda_k$. Gradients reach J_ϑ and R_ϑ and the graph-connected path/terminal losses, but not the Hessians of H_d or e. This is the closest extension of the repository default.
2. **Gain-only controller learning.** Detach L_k, e_k, and y_k as geometric features, but keep $u_k = u_\kappa(e_k, y_k)$ differentiable with respect to κ. This learns gains and schedules through the rollout without differentiating the spline-measurement Hessian.
3. **Full end-to-end mode.** Use `create_graph=True` for g_k, the measurement VJP that forms $L_k^\top\lambda_k$, and the JVP that forms y_k. This includes $A_\vartheta\nabla^2 H_d$, derivatives of $G = L^\top$ and u, and all control-point/knot mixed sensitivities. It is the most faithful mode and the most memory intensive.

Recommended automatic-differentiation order. At each rollout step: (i) evaluate H_d and g_k; (ii) sample the B-spline and evaluate the residual blocks in equation (129); (iii) compute $\lambda_k = \nabla_e\Psi$; (iv) obtain $L_k^\top\lambda_k$ by VJP and $y_k = L_k v_k$ by JVP; (v) assemble the momentum-row port force; and (vi) advance equation (155) while retaining the graph according to the selected mode. The terminal, path-potential, controller-storage, effort, and certificate-defect terms are then backpropagated through the complete rollout.

For long rollouts, checkpointing or truncated backpropagation may be necessary. Truncation changes the learned controller because it discards long-range sensitivity of an early geometry correction to the terminal curve; its horizon must therefore be reported. The knot controller should generally use a shorter active differentiation window or a slower refresh rate because $\partial_\gamma C$ and its mixed second derivatives are substantially less well conditioned than the control-point basis pullback.

B.7 Continuous energy certificate

Assume first that e has no explicit time dependence, the canonical kinematic relation holds, and $y = \dot{e}$. Using equations (118) and (149),

$$\begin{aligned} \frac{d}{dt}\big[H_d(x;Y) + \Psi(e)\big] &= -\nabla H_d^\top R_d \nabla H_d + y^\top(-\lambda - D_y y) + \lambda^\top \dot{e} \\ &= -\nabla H_d^\top R_d \nabla H_d - y^\top D_y y \le 0. \end{aligned} \tag{169}$$

The controller storage cancels the conservative port exchange, while R_d and D_y dissipate energy. This is the desired pH certificate.

For moving targets, externally changing measurements, or noncanonical learned kinematics, equation (146) gives the qualified balance

$$\frac{d}{dt}\big[H_d + \Psi\big] = -\nabla H_d^\top R_d \nabla H_d - y^\top D_y y + \lambda^\top \delta_e + \lambda^\top \partial_t e. \tag{170}$$

The last two terms are not numerical noise: they are measured power defects caused by kinematic mismatch and explicit measurement motion. They must be reported before claiming passivity of the learned closed loop.

If the controller uses clipping or projection, the certificate is preserved only when the implemented effort remains passive. A safer design is to obtain bounded efforts from a smooth storage with bounded gradient rather than clipping $L^\top\lambda$ after it has been computed. Knot trust-region projection should be accompanied by an acceptance test or by logging the projection work as an additional power term.

B.8 Separate control-point and knot controllers

The two ports should be implemented and tuned separately even when they are driven by related geometric measurements.

Control-point port. The fast controller acts every state step and handles measurements that require immediate geometric correction:

$$e_q = \begin{bmatrix} e_{\mathrm{nb}} \\ e_{\mathrm{xing}} \\ e_{\mathrm{orient}} \\ e_{\mathrm{sep}} \\ e_{\mathrm{strip}} \end{bmatrix}, \qquad G_q = \begin{bmatrix} 0 \\ 0 \\ L_q^\top \\ 0 \end{bmatrix}, \qquad u_q = -\nabla_{e_q}\Psi_q - D_q^{\mathrm{port}} y_q. \tag{171}$$

Because B-spline basis support is local, the VJP distributes each sample-space correction only to nearby control points. Use this port for target-field tracking, local orientation repair, collision anticipation, and strip or prism quality. Gains may be relatively high, but step rejection is still required for an exact segment crossing or negative strip Jacobian.

Knot port. The slow controller primarily handles parameterization and support distribution:

$$e_\gamma = \begin{bmatrix} e_{\text{interval floor}} \\ e_{\text{spacing}} \\ e_{\text{support}} \\ e_{\text{arc-length allocation}} \end{bmatrix}, \qquad G_\gamma = \begin{bmatrix} 0 \\ 0 \\ 0 \\ L_\gamma^\top \end{bmatrix}, \qquad u_\gamma = -\nabla_{e_\gamma}\Psi_\gamma - D_\gamma^{\mathrm{port}} y_\gamma. \tag{172}$$

Use $M_\gamma \gg M_q$, stronger damping, a warm-up period, and less frequent controller refreshes. The fixed-period chart should remain active so that the controller redistributes interval mass rather than changing the total period. A practical schedule is to update u_q every step and refresh u_γ every N_γ steps only when the geometric controller has stagnated or when interval/support diagnostics exceed thresholds.

A geometric residual generally depends on both q and γ. If it must drive both ports with an exact certificate, use one shared measurement e and the combined map

$$G_e = \begin{bmatrix} 0 \\ 0 \\ L_q^\top \\ L_\gamma^\top \end{bmatrix}, \qquad y_e = L_q v_q + L_\gamma v_\gamma, \tag{173}$$

with one effort $u_e = -\nabla_e\Psi - D_e y_e$. Arbitrarily scaling the two blocks of G_e changes the measured output and destroys the identity $y_e = \dot{e} - \partial_t e$. Time-scale separation should therefore be introduced through masses, damping, activation schedules, or separate controller storages, not by silently rescaling a measurement-aligned G.

Recommended measurement and port assignment.

Measurement	Preferred port	Interpretation
Target SDF or UDF/narrow-band residual	q; optionally shared	Current boundary motion relative to the observed target field.
LIVE/O&R crossing or orientation residual	q	Local span or sampled-window anti-degeneracy feedback; not a topology certificate.
Nonadjacent segment distance	q	Nonlocal collision-avoidance violation; supplement with exact crossing rejection.
Strip Jacobian or quality residual	q	Local foldover, inversion, and sliver feedback for a ribbon/prism representation.
Interval floor and interval ratio	γ	Safe knot redistribution under the fixed-period chart.
Arc-length/support-density mismatch	γ or shared	Reallocates parameter resolution; depends on both geometry and knots.
Global target divergence	plant potential	Nominal task energy; not by itself a topology controller.

B.9 Discrete certificate and required diagnostics

The continuous certificate does not survive an arbitrary explicit-Euler step. For a proposed update x_{k+1}, recompute

$$E_k = H_d(x_k; Y_k) + \Psi(e_k) \tag{174}$$

at the stored state, and form the predicted passive power

$$P_k^{\text{pred}} = -g_k^\top R_{d,k} g_k - y_k^\top D_{y,k} y_k, \qquad g_k = \nabla H_d(x_k; Y_k). \tag{175}$$

A controller-aware explicit step should be accepted only if

$$E_{k+1} - E_k \le h P_k^{\text{pred}} + h\lambda_k^\top(\delta_{e,k} + \partial_t e_k) + \eta_E h^2, \tag{176}$$

where $\eta_E h^2$ is a documented truncation tolerance. Backtracking or substepping is required when this test fails. A discrete-gradient implementation can replace equation (176) by an exact discrete power identity, but then both the plant and controller storages should use compatible discrete gradients.

The implementation should log, at every accepted step,

$$\begin{gathered} H_d, \quad \Psi, \quad E = H_d + \Psi, \quad y_q^\top u_q, \quad y_\gamma^\top u_\gamma, \\ g^\top R_d g, \quad y^\top D_y y, \quad \|\delta_e\|, \quad \frac{E_{k+1} - E_k}{h} - P_k^{\text{pred}}, \\ \|L_q^\top \lambda\|, \quad \|L_\gamma^\top \lambda\|, \quad \|e_{\text{nb}}\|, \ \|e_{\text{xing}}\|, \ \|e_{\text{sep}}\|, \quad \min_i \Delta_i, \end{gathered} \tag{177}$$

together with self-intersections and any hard safety events. These quantities distinguish plant dissipation, controller work, moving-measurement supply, and numerical integration error.

B.10 Minimal implementation sequence

A direct implementation can be organized as follows.

1. Select and record one differentiation mode: stop-gradient force, gain-only controller learning, or full end-to-end differentiation.
2. Evaluate the nominal plant energy H_d and its gradient $g = \nabla H_d$ with `create_graph` set consistently with that mode.
3. Sample the B-spline and evaluate UDF/narrow-band, LIVE/O&R, segment-separation, optional strip-quality, and knot-safety residual blocks without adding them to the nominal plant potential.
4. Compute controller storages and efforts $\lambda_q = \nabla\Psi_q(e_q)$ and $\lambda_\gamma = \nabla\Psi_\gamma(e_\gamma)$.
5. Use VJPs to obtain $L_q^\top \lambda_q$ and $L_\gamma^\top \lambda_\gamma$; use JVPs along (v_q, v_γ) to obtain y_q, y_γ. Retain or detach these graphs according to the selected mode.
6. Assemble $G_q u_q + G_\gamma u_\gamma$ in the momentum rows of the pH right-hand side and advance the controlled explicit step.

7. Accumulate the graph-connected terminal/path loss together with optional controller-storage, effort, and certificate-defect terms from equation (158).
8. Recompute H_d, Ψ, all measurements, and hard safety diagnostics at the proposed state; apply equation (176) and reject exact crossings, negative strip Jacobians, or invalid knot intervals.
9. Backpropagate through the accepted rollout. Update the control-point controller every accepted step; refresh the knot controller only after warm-up and at its slower scheduled rate.

Recommended first code change

Replace the current interpretation of all correction terms as either V or R_{ctrl} by three explicit interfaces: `measurement(state,target)` returning e; `controller(e,y)` returning u and controller storage Ψ; and `port(state,e)` providing VJP/JVP operations for Gu and $y = G^\top \nabla H_d$. Retain the safe mechanical kinematic block while validating the identity $y \approx \dot{e} - \partial_t e$. Only after this certificate is numerically verified should the port maps or controller gains be learned.

C Target-Driven Port-Hamiltonian Interpolation with Information-Geometric Conditioning

This appendix isolates the role of the information-geometric terms in the spline-deformation problem while retaining the port-Hamiltonian (pH) deformation model as the dynamical mechanism. The target divergence remains the term that drives the source spline toward the observed target. The Souriau–Fisher and measurement-Fisher tensors are used only to condition dissipative directions of the pH dynamics; they do not define a competing deformation flow and they do not replace the target discrepancy.

The geometric question considered here is whether the pH trajectory

$$t \longmapsto C(z_t; \cdot), \qquad z_t = (q_t, \gamma_t), \tag{178}$$

can interpolate from the initial circle toward the observed crescent while remaining a smooth family of embedded closed curves and retaining the intended SE(2) equivariance. Numerically, this is tested directly on the evolving spline. An auxiliary RKHS ambient flow is not used as the dynamics in this appendix.

C.1 Relation to the port-controller construction in Appendix B

Appendix B emphasizes an explicit measurement port

$$y_e = L(z)\dot{z}, \qquad u_e = -\nabla_e \Psi(e) - D_e y_e, \qquad f_e = L(z)^\top u_e, \tag{179}$$

and studies the corresponding power exchange between a pH plant and a controller storage. The present appendix uses the same pH viewpoint but asks a narrower numerical question. The plant Hamiltonian contains the target-matching divergence, while the information tensors enter the dissipative operator of the plant or an equivalent static output-damping closure. No inverse-Fisher natural-gradient flow is introduced as a second deformation law.

In the special case in which Appendix B uses a static output-damping controller and the canonical spline kinematics hold, its damping force and the metric damping used below are equivalent after pulling the measurement tensor back to the configuration velocity. The distinction here is organizational: the target potential, pH dynamics, information damping, and geometric path certificate are kept as separate numerical objects.

C.2 Spline configuration, phase space, and observed target

Let

$$z = (q, \gamma) \in \mathcal{Q}_{\mathrm{spl}}, \qquad q = (q_1, \ldots, q_n) \in (\mathbb{R}^2)^n, \qquad \gamma \in \mathbb{R}^m \tag{180}$$

denote the spline configuration. The corresponding phase-space state is

$$x = (z, p) = (q, \gamma, p_q, p_\gamma) \in T^* \mathcal{Q}_{\mathrm{spl}}. \tag{181}$$

The common-shift gauge of the knot logits is removed by

$$P_\gamma = I_m - \frac{1}{m}\mathbf{1}\mathbf{1}^\top, \qquad \gamma \leftarrow P_\gamma \gamma, \tag{182}$$

and the physical knot intervals are reconstructed through the fixed-period chart

$$\Delta_i(\gamma) = \delta + (P - m\delta)\frac{\exp(\gamma_i)}{\sum_{j=1}^m \exp(\gamma_j)}, \qquad \Delta_i > 0, \qquad \sum_i \Delta_i = P. \tag{183}$$

Let $u_1, \ldots, u_N$ be fixed material parameter samples on S^1 and define the observed current boundary

$$X(z) = \{X_a(z)\}_{a=1}^N, \qquad X_a(z) = C(q, \gamma; u_a). \tag{184}$$

The target is an observed boundary sample

$$Y = \{y_b\}_{b=1}^{N_Y} \subset \mathbb{R}^2, \qquad \nu_Y = \frac{1}{N_Y}\sum_{b=1}^{N_Y} \delta_{y_b}. \tag{185}$$

For the circle-to-crescent experiment, Y is stored as an ordered crescent boundary or as the same set viewed as an empirical measure. It is not interpreted as a periodic spline control polygon.

C.3 The divergence remains the target-driving potential

Let

$$\mu_z = \frac{1}{N}\sum_{a=1}^N \delta_{X_a(z)} \tag{186}$$

be the empirical measure induced by the sampled spline. The target potential is

$$V_{\mathrm{match}}(z; Y) = \lambda_Y \, \mathcal{D}(\mu_z, \nu_Y), \tag{187}$$

where $\mathcal{D}$ is fixed within an individual ablation. The pH model may additionally contain a representation or smoothness term $V_{\mathrm{reg}}(z)$, but the information tensors introduced below are not added to V_{match}.

The implementation supports multiple target discrepancies while retaining the same state and pH integrator. Representative choices are:

- an entropic optimal-transport cost or Sinkhorn divergence, with the same entropic temperature and iteration count across a metric ablation;
- sliced Wasserstein distance, with a fixed projection set or fixed projection seed;
- a symmetric point-set discrepancy such as Chamfer distance for diagnostic smoke tests;
- a kernel discrepancy such as MMD; and
- a Rényi/free-energy target term when the corresponding normalized measure model is used.

For each choice, the gradient used by the pH dynamics is obtained by differentiating the scalar target potential with respect to the spline configuration,

$$g_z = \nabla_z V_{\mathrm{match}}(z; Y) = \left(\frac{\partial X}{\partial z}\right)^\top \nabla_X \big[\lambda_Y \mathcal{D}(\mu_X, \nu_Y)\big]. \tag{188}$$

Thus the target force is always a pullback of an observation-space discrepancy, even though the state evolution is computed in the spline phase space.

C.4 Port-Hamiltonian deformation template

Define the shaped Hamiltonian

$$H_d(x; Y) = \frac{1}{2}p^\top M^{-1} p + V_{\mathrm{match}}(z; Y) + V_{\mathrm{reg}}(z), \tag{189}$$

where $M \succ 0$ is the configuration mass matrix. For the core numerical test, the kinematic block is kept canonical and any learned pH structure is placed in the momentum subsystem:

$$\begin{bmatrix} \dot{z} \\ \dot{p} \end{bmatrix} = \left[\begin{array}{cc} 0 & I \\ -I & J_{p,\theta}(x) \end{array} - \begin{array}{cc} 0 & 0 \\ 0 & R_{p,\theta}(x) + D_{\text{info}}(x) \end{array} \right] \begin{bmatrix} \nabla_z H_d \\ \nabla_p H_d \end{bmatrix}. \tag{190}$$

The learned block satisfies

$$J_{p,\theta}^{\top} = -J_{p,\theta}, \qquad R_{p,\theta} = R_{p,\theta}^{\top} \succeq 0. \tag{191}$$

The canonical configuration velocity is therefore

$$v_z := \dot{z} = M^{-1} p, \tag{192}$$

and the momentum equation can be read as

$$\dot{p} = -\nabla_z \big(V_{\text{match}} + V_{\text{reg}} \big) + J_{p,\theta} v_z - \big(R_{p,\theta} + D_{\text{info}} \big) v_z. \tag{193}$$

This form makes the roles of the terms explicit: the divergence supplies target attraction, the skew block supplies power-preserving momentum routing, and the positive-semidefinite blocks supply dissipation.

For a fixed target and deterministic dynamics,

$$\dot{H}_d = -v_z^{\top} \big(R_{p,\theta} + D_{\text{info}} \big) v_z \leq 0. \tag{194}$$

The divergence itself need not decrease monotonically at every instant because kinetic and potential energy can exchange through the Hamiltonian dynamics. The total Hamiltonian is the appropriate pH certificate, while V_{match} is recorded separately as the task error.

C.5 SE(2) pose directions and horizontal spline directions

The SE(2) action is diagonal on the control polygon and trivial on the knot logits. With $c(q)$ denoting the control-point centroid, define the centered generator

$$B_q(q) = \begin{bmatrix} J_2(q_1 - c(q)) & I_2 \\ \vdots & \vdots \\ J_2(q_n - c(q)) & I_2 \end{bmatrix}, \qquad J_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \tag{195}$$

The full generator on $z = (q, \gamma)$ is

$$B_G(z) = \begin{bmatrix} B_q(q) \\ 0_{m \times 3} \end{bmatrix} \in \mathbb{R}^{(2n+m) \times 3}. \tag{196}$$

Given a reference configuration metric $M_z \succ 0$, define the metric left inverse

$$B_G^{\sharp} = \big(B_G^{\top} M_z B_G \big)^{-1} B_G^{\top} M_z. \tag{197}$$

The reduced rigid velocity is

$$\nu_G = B_G^{\sharp} v_z. \tag{198}$$

Let Z_H be an M_z-orthonormal basis of the complement of the SE(2) orbit and the knot common-shift gauge. The horizontal velocity is

$$\nu_H = Z_H^{\sharp} v_z, \qquad Z_H^{\sharp} = Z_H^{\top} M_z, \tag{199}$$

and locally

$$v_z = B_G \nu_G + Z_H \nu_H. \tag{200}$$

The basis is recomputed after any active-set restriction so that masking of control points does not reintroduce rigid or knot-gauge directions.

C.6 Measurement Fisher block

Let

$$e(z) = \begin{bmatrix} e_{\mathrm{sep}}(z) \\ e_{\mathrm{turn}}(z) \\ e_{\mathrm{edge}}(z) \\ e_{\mathrm{area}}(z) \\ e_{\mathrm{knot}}(z) \end{bmatrix} \tag{201}$$

collect internal geometric measurements. The target discrepancy is deliberately excluded from this stack. With

$$L(z) = \frac{\partial e}{\partial z}, \qquad W \succeq 0, \tag{202}$$

the configuration-space measurement tensor is

$$\mathcal{I}_z = L^\top W L. \tag{203}$$

The reduced horizontal tensor is

$$\mathcal{G}_H = Z_H^\top \mathcal{I}_z Z_H + \lambda_H M_H, \qquad \lambda_H > 0. \tag{204}$$

Rigid-motion invariance of the internal measurements is checked numerically through

$$\epsilon_{LB} = \frac{\|LB_G\|_F}{\|L\|_F \|B_G\|_F + \varepsilon_0}, \tag{205}$$

and the knot-gauge residual is checked by applying L to the common-shift direction.

C.7 Souriau–Fisher pose block

For the centered SE(2) generator, use the centered momentum statistic

$$\mathbf{J}_c(x) = \begin{bmatrix} \sum_{i=1}^{n} (q_i - c(q)) \times p_{q_i} \\ \sum_{i=1}^{n} p_{q_i,x} \\ \sum_{i=1}^{n} p_{q_i,y} \end{bmatrix} \in \mathfrak{se}(2)^*. \tag{206}$$

Given an ensemble $\{x^{(b)}\}_{b=1}^{B}$ with normalized nonnegative weights w_b, define

$$\widehat{Q} = \sum_b w_b \mathbf{J}_c(x^{(b)}), \qquad \widehat{\mathcal{I}}_S = \sum_b w_b \big(\mathbf{J}_c(x^{(b)}) - \widehat{Q}\big)\big(\mathbf{J}_c(x^{(b)}) - \widehat{Q}\big)^\top. \tag{207}$$

The regularized pose tensor is

$$\mathcal{G}_S = (1 - \rho_S)\widehat{\mathcal{I}}_S + \rho_S \frac{\mathrm{tr}(\widehat{\mathcal{I}}_S)}{3} I_3 + \lambda_S M_S. \tag{208}$$

The ensemble is fixed over the inner rollout or updated only at a documented outer stage. Its eigenvalues, rank, and condition number are stored before the tensor is used in the dynamics.

An observation-space pose tensor can be monitored separately by sampling the current spline and evaluating the infinitesimal rigid generators on the observed points. This auxiliary tensor diagnoses whether the observed geometry contains enough rotational and translational information; it is not substituted for V_{match}.

C.8 Lifting the information metrics into pH dissipation

The information damping force is assembled from reduced pose and horizontal velocities. Let

$$D_G = \alpha_G \mathcal{G}_S, \qquad D_H = \alpha_H \mathcal{G}_H, \qquad \alpha_G, \alpha_H \geq 0. \tag{209}$$

The corresponding generalized damping operator and force are

$$D_{\text{info}} = (B_G^\sharp)^\top D_G B_G^\sharp + (Z_H^\sharp)^\top D_H Z_H^\sharp, \tag{210}$$

$$f_{\text{info}} = -D_{\text{info}} v_z = -(B_G^\sharp)^\top D_G \nu_G - (Z_H^\sharp)^\top D_H \nu_H. \tag{211}$$

Its dissipated power is

$$v_z^\top f_{\text{info}} = -\nu_G^\top D_G \nu_G - \nu_H^\top D_H \nu_H \le 0. \tag{212}$$

This is the numerical meaning of the information terms in the core pH experiment. They alter the anisotropy of dissipation but they do not create a new target potential and they do not invert the Fisher tensor to define a natural-gradient state update.

The four primary metric variants are

$$(\alpha_G, \alpha_H) \in \{(0,0), (0,+), (+,0), (+,+)\}, \tag{213}$$

corresponding to no information metric, horizontal measurement FIM only, Souriau pose metric only, and both blocks.

C.9 Diffeomorphic path criterion through spline isotopy

The pH dynamics evolve the spline configuration directly. Diffeomorphic admissibility is therefore tested on the curve path rather than imposed by a separate ambient-flow model. Define

$$F : S^1 \times [0,T] \to \mathbb{R}^2, \qquad F(u,t) = C(z_t; u). \tag{214}$$

A continuous-time trajectory for which every map $F_t = F(\cdot, t)$ is a smooth embedding gives an isotopy of the initial embedded circle. Such an isotopy admits an ambient extension to a family of plane diffeomorphisms. The numerical experiment therefore checks whether the computed spline path remains inside the embedding set.

For a dense material grid u_a, the following quantities are monitored at every accepted step and at the integrator midpoint:

$$s_{\min,k} = \min_a \|\partial_u C(z_k; u_a)\|, \tag{215}$$

$$d_{\text{sep},k} = \min_{(a,b)\in\mathcal{N}_{\text{nonadj}}} \|X_{a,k} - X_{b,k}\|, \tag{216}$$

$$N_{\text{cross},k} = \text{number of nonadjacent segment intersections}, \tag{217}$$

$$A_k = \frac{1}{2} \sum_a \big(X_{a,k} \times X_{a+1,k}\big). \tag{218}$$

A proposed step is geometrically admissible when

$$s_{\min,k} > \varepsilon_{\tan}, \qquad d_{\text{sep},k} > \varepsilon_{\text{sep}}, \qquad N_{\text{cross},k} = 0, \qquad \operatorname{sign}(A_k) = \operatorname{sign}(A_0), \tag{219}$$

and all knot intervals remain above their prescribed floor. The same checks are applied to midpoint/substep states. Failure triggers time-step subdivision rather than adding a new target-independent potential.

The resulting discrete test is a numerical certificate of a resolved embedding isotopy. It is stronger than checking only the final self-intersection count because it tests the complete computed path, while the ambient diffeomorphism is inferred from the embedding isotopy rather than generated by a second dynamics.

C.10 SE(2) symmetry tests and randomly rotated crescent targets

Two different experiments are required.

First, structural equivariance is tested by applying the same rigid transformation $g \in \mathrm{SE}(2)$ to the source and target and comparing

$$\epsilon_{\text{eq}} = \frac{\|\Phi_T(g\cdot x_0, g\cdot Y) - g\cdot \Phi_T(x_0, Y)\|}{\|g\cdot \Phi_T(x_0,Y)\| + \varepsilon_0}. \tag{220}$$

When the Souriau ensemble is transformed for this test, both positions and momenta are transformed consistently with the rigid frame change.

Second, pose sensitivity is tested by leaving the source circle fixed and rotating the target crescent about its centroid. Let

$$Y_\vartheta = c_Y + R(\vartheta)(Y - c_Y), \qquad \vartheta \sim \mathrm{Unif}[-\vartheta_{\max}, \vartheta_{\max}]. \tag{221}$$

The same sampled angle ϑ is used across the four information-metric variants of a trial. This changes the task rather than the coordinate frame and therefore measures whether pose-conditioned dissipation changes robustness to target orientation.

For each trial, record both the raw observation discrepancy and the discrepancy after a best-fit rigid alignment. The difference separates global pose error from intrinsic shape error without altering the target energy used by the pH dynamics.

C.11 Semi-implicit pH time integration

Let

$$D_k = R_{p,\theta}(x_k) + D_{\mathrm{info}}(x_k) \succeq 0, \qquad J_k = J_{p,\theta}(x_k). \tag{222}$$

The implementation evaluates

$$g_k = \nabla_z \left[V_{\mathrm{match}}(z_k; Y) + V_{\mathrm{reg}}(z_k)\right] \tag{223}$$

and uses an implicit treatment of the dissipative momentum term,

$$\left(I + hD_k M^{-1}\right) p_{k+1} = p_k + h\left[-g_k + J_k M^{-1} p_k\right]. \tag{224}$$

The configuration is then updated by

$$z_{k+1} = z_k + hM^{-1} p_{k+1}, \tag{225}$$

followed by the knot-gauge projection and reconstruction of positive physical knot intervals. Equation (224) requires only a positive-definite linear solve after the structured pH blocks have been evaluated.

The candidate state is accepted only after the embedding-isotopy tests in equation (219) have also been evaluated at the midpoint state

$$z_{k+1/2} = \frac{1}{2}(z_k + z_{k+1}) \tag{226}$$

or through an equivalent half-step reconstruction. If the geometric gate or energy certificate fails, the same pH step is recomputed with $h \leftarrow h/2$.

For the deterministic fixed-target problem, the discrete power defect is monitored as

$$r_{H,k} = H_d(x_{k+1}; Y) - H_d(x_k; Y) + h\, v_{z,k+1}^\top D_k v_{z,k+1}. \tag{227}$$

A truncation-scaled acceptance condition is

$$r_{H,k} \le \eta_H h^2. \tag{228}$$

The target divergence V_{match} is logged independently because a pH trajectory can exchange kinetic and target potential energy while still satisfying the total-energy inequality.

C.12 Differentiation and implementation of the divergence force

The numerical implementation evaluates V_{match} from sampled spline points, but the pH state remains $(q, \gamma, p_q, p_\gamma)$. The required generalized force is computed by reverse-mode differentiation through the spline sampler and the selected divergence:

(i) reconstruct the physical intervals from γ_k;
(ii) evaluate $X_k = C(q_k, \gamma_k; u_{1:N})$;
(iii) evaluate $\mathcal{D}(\mu_{X_k}, \nu_Y)$ using the configured divergence;
(iv) differentiate the scalar result with respect to (q_k, γ_k) to obtain g_k in equation (223);
(v) remove the common-shift component of the knot gradient by P_γ before the momentum update.

For Sinkhorn-type terms, the iteration count and entropic temperature are treated as part of the divergence definition and are fixed during a metric ablation. If differentiation through a finite number of Sinkhorn iterations is used, that finite-iteration gradient is the force actually integrated by the pH solver and must therefore be held fixed across variants.

The measurement Jacobian L need not be materialized densely when the number of measurements is large. The horizontal Fisher action can be assembled from Jacobian–vector and vector–Jacobian products. Dense matrices are retained in the small 2D validation problem because they permit direct eigenvalue, rank, and nullspace diagnostics.

C.13 Metric estimation and numerical regularization

The Souriau covariance is estimated from a stage-level ensemble rather than from a single instantaneous state. For rotated-crescent experiments, the ensemble can be assembled from the same training/evaluation task family, but the covariance used inside a rollout is frozen so that changes in the metric are not confounded with changes in the state integrator. The stored diagnostics include

$$\lambda_{\min}(\mathcal{G}_S), \quad \lambda_{\max}(\mathcal{G}_S), \quad \kappa(\mathcal{G}_S), \quad \operatorname{rank}(\widehat{\mathcal{I}}_S). \tag{229}$$

The same quantities are recorded for $\mathcal{G}_H$.

The regularization parameters λ_S and λ_H are not used to create target attraction. Their numerical role is to keep the reduced damping tensors well conditioned when the available measurements or momentum ensemble do not excite every reduced direction. Direct Cholesky factorizations are sufficient for the present 2D problem; iterative solves can replace them when the horizontal dimension becomes large.

C.14 Stochastic pH extension

When stochastic deformation is enabled, the deterministic pH drift remains equation (190). A Stratonovich perturbation is added to the momentum equation,

$$dz_t = M^{-1}p_t\,dt, \qquad dp_t = [-\nabla_z V + J_{p,\theta}v_z - Dv_z]\,dt + \Sigma_p(x_t) \circ dW_t. \tag{230}$$

The stochastic term is therefore applied to the pH momentum state rather than independently perturbing knot logits or control points in observation space. Observation noise is represented separately through the measurement model used to construct $\mathcal{D}$ or the measurement covariance. The deterministic and stochastic runs use the same target divergence and the same geometric embedding checks.

For a stochastic integrator, the embedding gate is evaluated after each realized proposal and at the available predictor state. A rejected step is recomputed with a smaller deterministic time increment while retaining a documented treatment of the Wiener increment so that the rejection rule is not confused with an exact invariant-measure sampler.

C.15 Numerical quantities recorded in each ablation

For each accepted step, the implementation records

$$\begin{gathered} H_d, \quad T = \frac{1}{2}p^\top M^{-1}p, \quad V_{\text{match}}, \quad V_{\text{reg}}, \quad r_H, \\ \|\nu_G\|, \quad \|\nu_H\|, \quad \nu_G^\top D_G \nu_G, \quad \nu_H^\top D_H \nu_H, \\ \lambda_i(\mathcal{G}_S), \quad \lambda_i(\mathcal{G}_H), \quad \epsilon_{LB}, \\ s_{\min}, \quad d_{\text{sep}}, \quad N_{\text{cross}}, \quad \operatorname{sign}(A), \quad h_k, \quad N_{\text{backtrack}}. \end{gathered} \tag{231}$$

The rotated-target experiments additionally record the sampled target angle, raw observation discrepancy, rigid-aligned observation discrepancy, and a pose-orientation error whenever the observed shape has a well-defined principal axis.

The projected phase plots use the pH state rather than an auxiliary flow coefficient. Representative projections are

$$(c_x(q), P_x), \qquad (\theta(q), L_c), \qquad (V_{\text{match}}, \|p_H\|)\,, \tag{232}$$

where P_x is total linear momentum, L_c is centered angular momentum, and p_H is the horizontal component of the momentum under the same pose/horizontal decomposition used by the metric blocks.

C.16 Self-contained end-to-end algorithm

Algorithm C.1: target-driven pH spline interpolation with information-metric ablation

Inputs. Initial spline state $x_0 = (q_0, \gamma_0, p_{q,0}, p_{\gamma,0})$; ordered source material parameters $u_{1:N}$; observed target boundary Y; divergence $\mathcal{D}$ and all of its numerical parameters; pH template parameters θ; mass matrix M; base pH damping; measurement definition $e(z)$ and precision W; Souriau ensemble; information variant in {none, horizontal, pose, both}; time step h; rollout length K; knot floor; embedding thresholds $\varepsilon_{\tan}$ and ε_{sep}; energy-defect tolerance η_H; and maximum number of step subdivisions.

Trial initialization.

1. If a pose-variation trial is requested, draw or load one angle ϑ and construct the rotated target Y_ϑ using equation (221). Use the same Y_ϑ for every metric variant of that trial.
2. Canonicalize γ_0 with equation (182), reconstruct positive knot intervals, and sample $X_0 = C(q_0, \gamma_0; u_{1:N})$.
3. Assemble the stage-frozen Souriau tensor $\mathcal{G}_S$ from equations (207) and (208). Record its eigenvalues, numerical rank, and condition number.
4. Set $x \leftarrow x_0$ and initialize the trajectory, energy, metric, phase-map, and embedding diagnostic buffers.

For $k = 0, \ldots, K-1$**:**

1. Reconstruct the physical knot intervals from γ_k and evaluate the sampled spline $X_k = C(q_k, \gamma_k; u_{1:N})$.
2. Evaluate the configured target potential

$$V_{\text{match},k} = \lambda_Y \mathcal{D}(\mu_{X_k}, \nu_Y)$$

   and obtain its generalized gradient with respect to (q_k, γ_k) by reverse-mode differentiation through the spline sampler and divergence computation. Project the knot-gradient component with P_γ.
3. Evaluate V_{reg}, the Hamiltonian equation (189), the learned structured blocks $J_{p,\theta}$ and $R_{p,\theta}$, and the current configuration velocity $v_{z,k} = M^{-1} p_k$.
4. Construct B_G, $B_G^\sharp$, and Z_H. Evaluate the internal measurements $e(z_k)$ and their Jacobian L_k. Assemble $\mathcal{G}_H$ from equation (204) and compute the rigid-invariance and knot-gauge residuals.
5. According to the selected metric variant, set D_G and D_H in equation (209); form the information force equation (211) and the total dissipative operator $D_k = R_{p,\theta} + D_{\text{info}}$.
6. Compute the semi-implicit pH proposal by solving

$$\left(I + h_k D_k M^{-1}\right) p^+ = p_k + h_k \left[-\nabla_z (V_{\text{match}} + V_{\text{reg}}) + J_{p,\theta} M^{-1} p_k\right],$$

   followed by

$$z^+ = z_k + h_k M^{-1} p^+.$$

   Apply the knot-gauge projection and reconstruct the physical intervals.
7. Evaluate the midpoint/half-step spline and the endpoint spline on a dense material grid. At both states compute the minimum tangent norm, nonadjacent separation, self-intersection count, signed-area orientation, and minimum knot interval. Reject the proposal if any condition in equation (219) fails.
8. Evaluate $H_d(x^+; Y)$ and the discrete energy residual $r_{H,k}$ in equation (227). Reject the proposal if equation (228) fails. For a rejected proposal, set $h_k \leftarrow h_k/2$ and repeat Steps 6–8 without changing the target, divergence, pH template, or metric variant.
9. Accept $x_{k+1} \leftarrow x^+$. Store H_d, kinetic energy, V_{match}, V_{reg}, r_H, pose/horizontal velocity norms and dissipated powers, metric spectra, phase-map coordinates, embedding diagnostics, accepted step size, and backtracking count.

After the rollout.

1. Compute the final raw target discrepancy and the rigid-aligned observation discrepancy. For rotated-target trials, store the sampled angle and the final pose-orientation error when defined.
2. Run the simultaneous source–target frame transformation test and compute the equivariance defect equation (220), transforming the Souriau momentum ensemble consistently with the frame change.
3. Repeat the complete rollout with the same initial state, target, divergence parameters, pH template, time-step policy, and embedding thresholds for the four variants in equation (213).
4. For a divergence ablation, repeat the same four metric variants after changing only $\mathcal{D}$ and its explicitly documented numerical parameters.